TRACE-HEP

Comprehensive Documentation - v0.1.13 - Toolkit for Rendering and Analysis of Collider Events

What is TRACE-HEP?

* A format-agnostic Python library for reconstructed-level collider event and vertex displays

* Polar (phi, |eta|), beam-axis 2D, and interactive 3D event views
* A companion per-vertex / all-vertices family for pileup diagnostics
* Core package has zero ROOT/uproot dependency
* Optional loaders for Delphes, flat-ntuple, calo-timing, and public
* ATLAS/CMS Open Data ROOT files
* A browsable HTML gallery for failure-mode/anomaly review at scale
* trace-gui: a live interactive viewer, incl. build-by-hand events
* Open source, MIT licensed

Where to find it

```
GitHub:    github.com/wasikatcern/TRACE-HEP
TestPyPI:  test.pypi.org/project/trace-hep
License:   MIT
```

Installation: core package

```
python3 -m venv .venv
source .venv/bin/activate

pip install --index-url https://test.pypi.org/simple/ \
            --extra-index-url https://pypi.org/simple/ \
                              trace-hep
```

Core install pulls only numpy, matplotlib, and plotly.

Installation: optional extras

```
# ROOT-file loaders: Delphes, flat ntuple, calo-timing
pip install --index-url https://test.pypi.org/simple/ \
            --extra-index-url https://pypi.org/simple/ \
            "trace-hep[delphes]"

# + static image export (kaleido) and ATLAS style (mplhep)
pip install ... "trace-hep[all]"

# from source instead, editable (for developing tracehep itself)
git clone https://github.com/wasikatcern/TRACE-HEP.git
cd TRACE-HEP
pip install -e ".[dev]"
pytest -q
```

Verify the install

```
python3 -c "import tracehep as trace; print(trace.__version__)"
# -> 0.1.13

trace-batch --help
# -> usage: trace-batch [-h] --input FILE.root --indices N [N ...] ...
```

The data model: one hard-scatter event

```
Jet(pt, eta, phi, mass=0, btag=False, is_hs=None)
Track(pt, eta, phi, charge=0, d0=0, z0=0, x=0, y=0, z=0,
      time=None, is_hs=None)
Lepton(pt, eta, phi, flavor="muon"|"electron")
Photon(pt, eta, phi)
MissingET(pt, phi, eta=0)

Event(jets, tracks, muons, electrons, photons, met,
      run=None, event_number=None, label="")
```

The data model: one pileup scenario

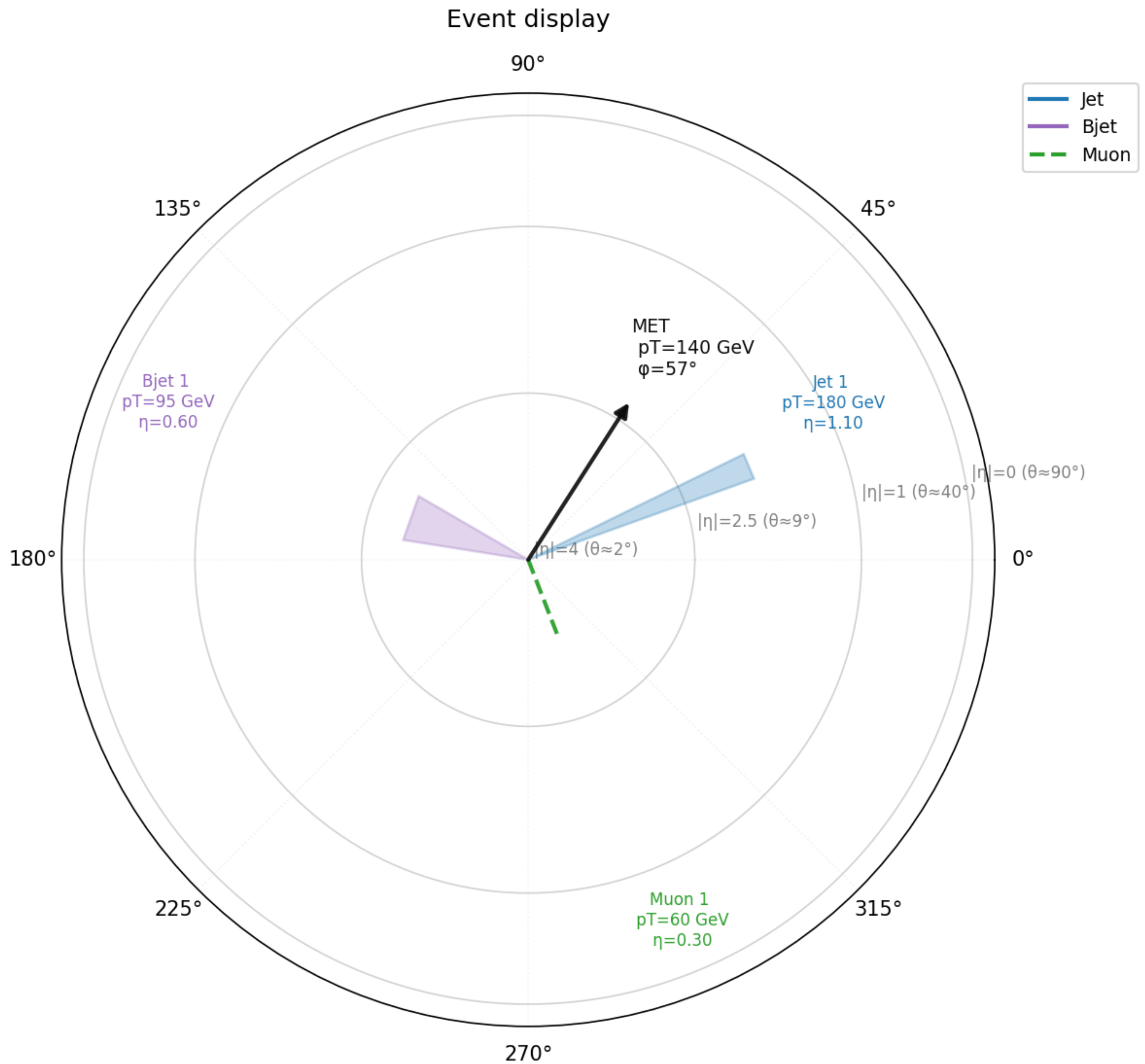
```
Vertex(z, x=0, y=0, sum_pt2=0, is_hs=False, is_pu=False,
       time=None, track_indices=[])
TruthVertex(z, x=0, y=0, is_hs=False)

VertexEvent(vertices, tracks, truth_vertices, mu=None, label="")
```

Nothing here knows about ROOT or any ntuple format -- build these by hand from your own analysis objects and every drawing function in the package works unchanged.

Quickstart: build an event by hand

```
import tracehep as trace

event = trace.Event(
jets=[trace.Jet(pt=180, eta=1.1, phi=0.4),
trace.Jet(pt=95, eta=-0.6, phi=2.8, btag=True)],
muons=[trace.Lepton(pt=60, eta=0.3, phi=-1.2, flavor="muon")],
met=trace.MissingET(pt=140, phi=1.0),
)

fig = trace.plot_event_polar(event)
fig.savefig("event_polar.png")
```

Output: the polar (phi, |eta|) view



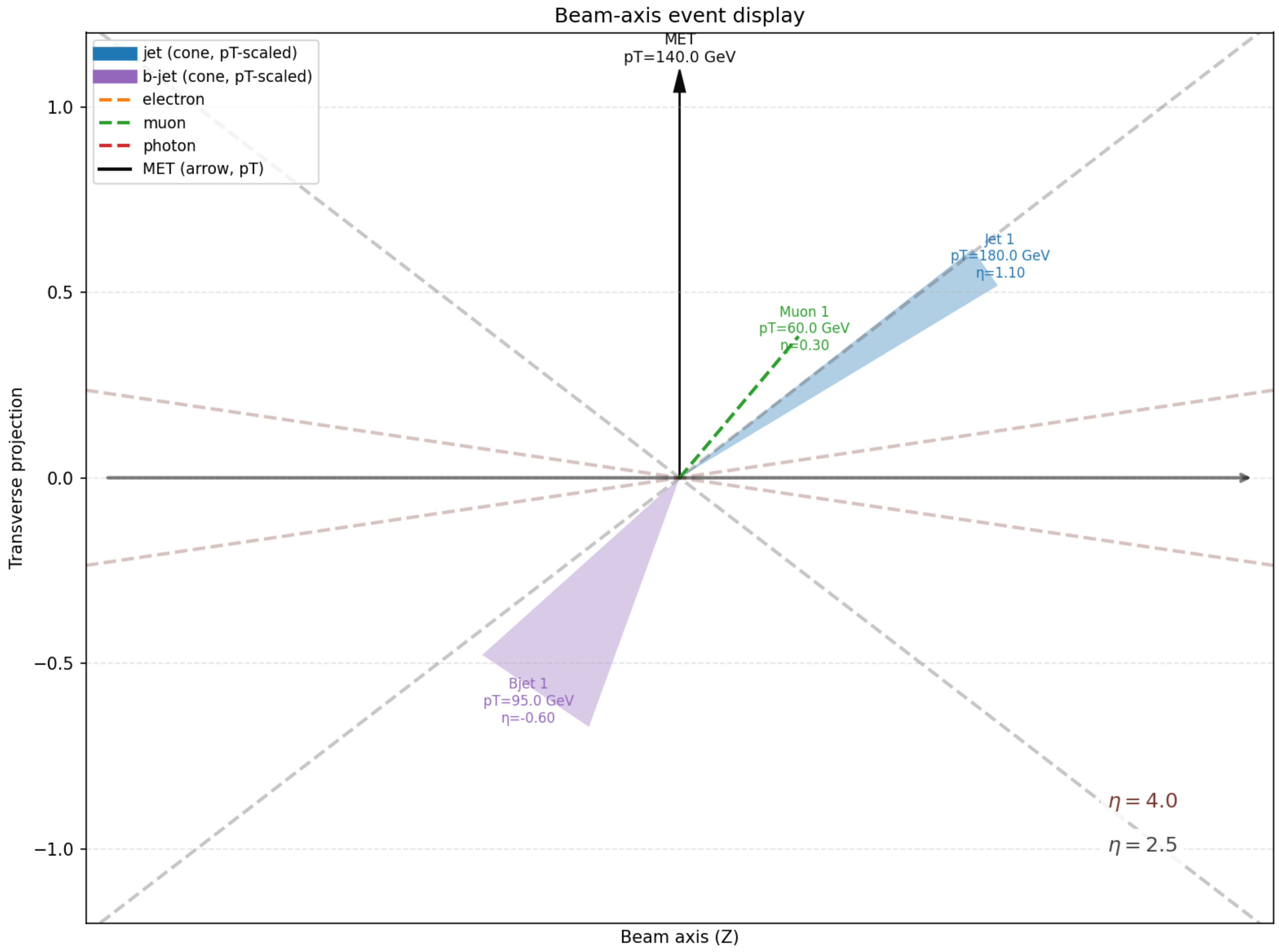
trace.plot_event_polar(event)

The beam-axis 2D projection

```
fig = trace.plot_event_beam2d(event)
fig.savefig("event_beam2d.png")
```

Complementary side-on view: beam direction horizontal, transverse coordinate vertical.

Output: the beam-axis 2D view

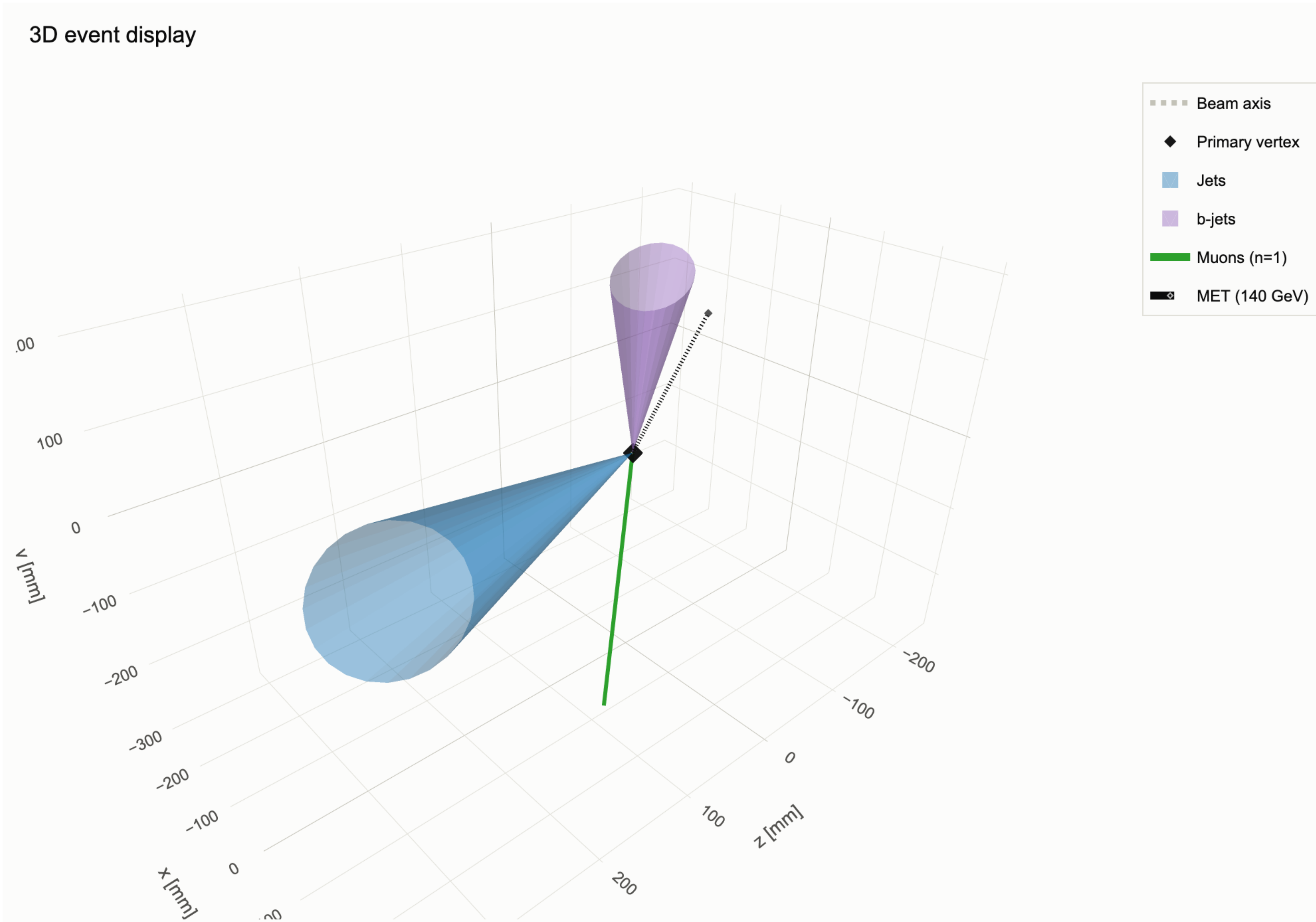


trace.plot_event_beam2d(event)

The interactive 3D view

```
fig3d = trace.plot_event_3d(event)
fig3d.write_html("event_3d.html")

# static snapshot (needs the kaleido extra)
fig3d.write_image("event_3d.png")
```

Output: the interactive 3D view



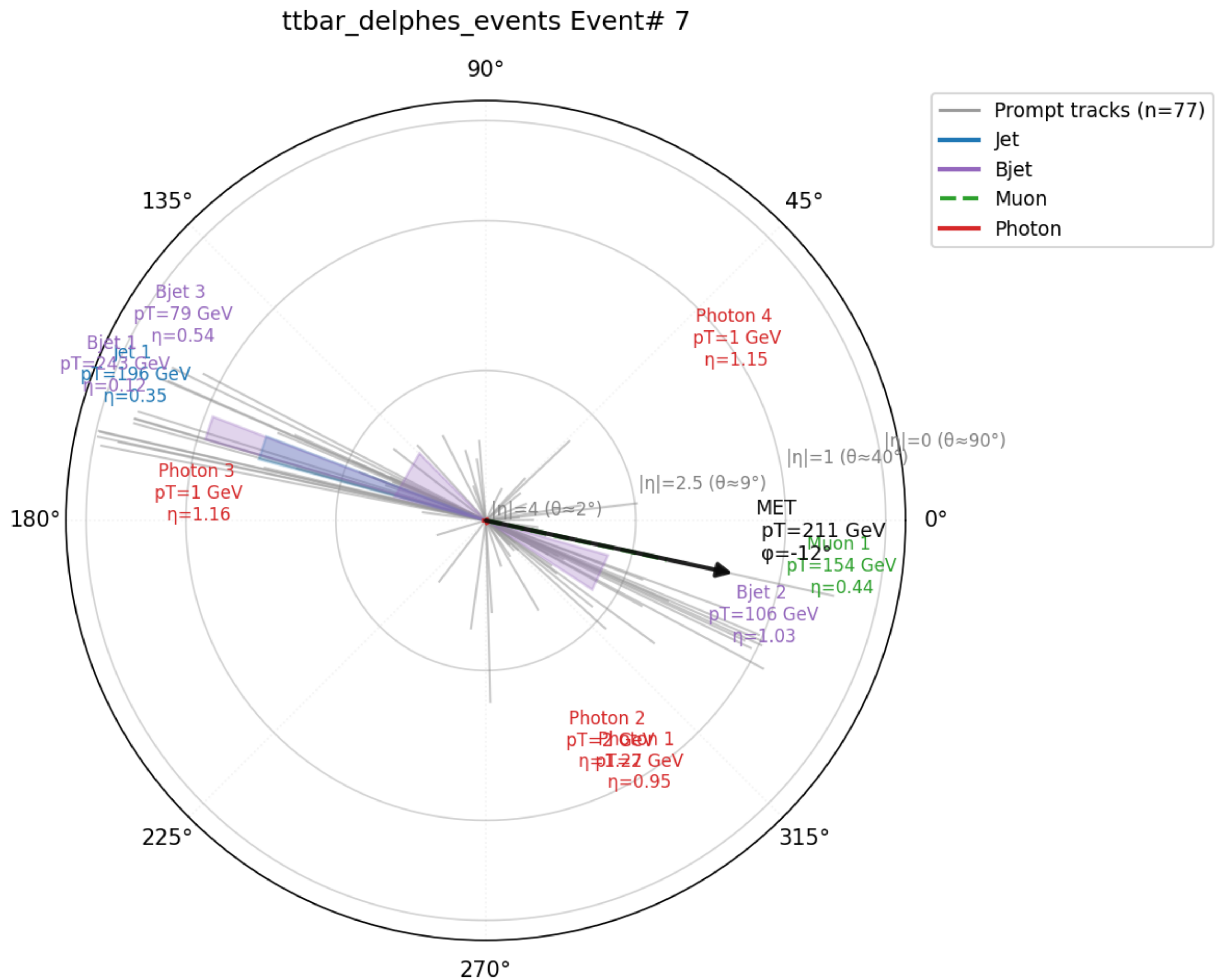
Static snapshot of the interactive HTML (drag to orbit, scroll to zoom)

Loading real events: Delphes ROOT files

```
from tracehep.io.delphes import load_events

events = load_events(
    "ttbar_delphes_events.root",
    indices=[7],
    label="ttbar_delphes_events",
)
event = events[7]

fig = trace.plot_event_polar(event, show_tracks=True)
fig.savefig("event7_polar.png")
```

Needs the [delphes] extra. Works unchanged on any Delphes sample, any physics process.

Output: a real ttbar Delphes event



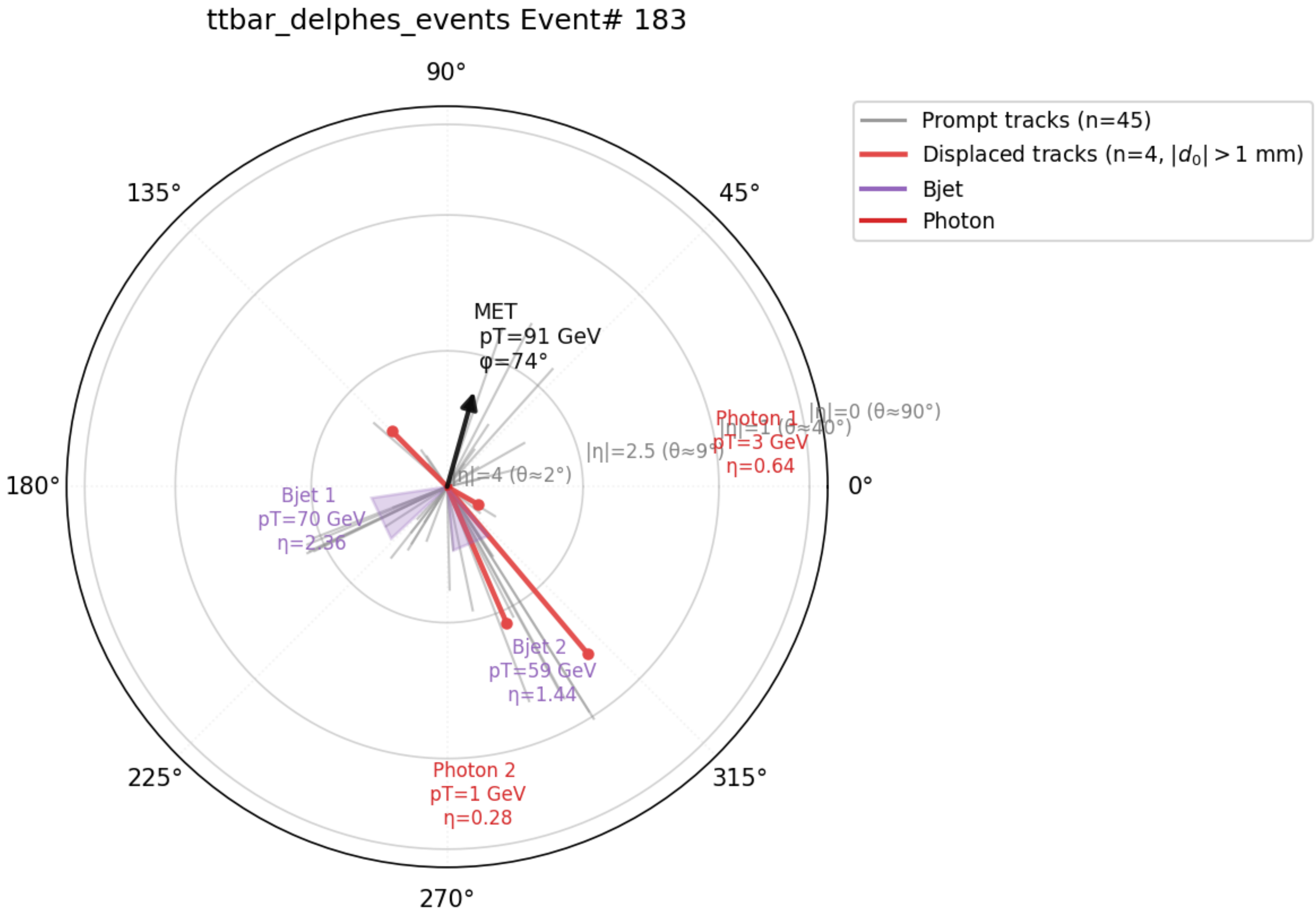
Event #7: 4 jets, 77 tracks, 1 muon, 4 photons, MET = 211 GeV

Highlighting displaced tracks (d0)

```
fig = trace.plot_event_polar(
    event, show_tracks=True, d0_displaced_mm=1.0,
)
fig.savefig("event_displaced_polar.png")

fig3d = trace.plot_event_3d(event, d0_displaced_mm=1.0)
fig3d.write_html("event_displaced_3d.html")
```

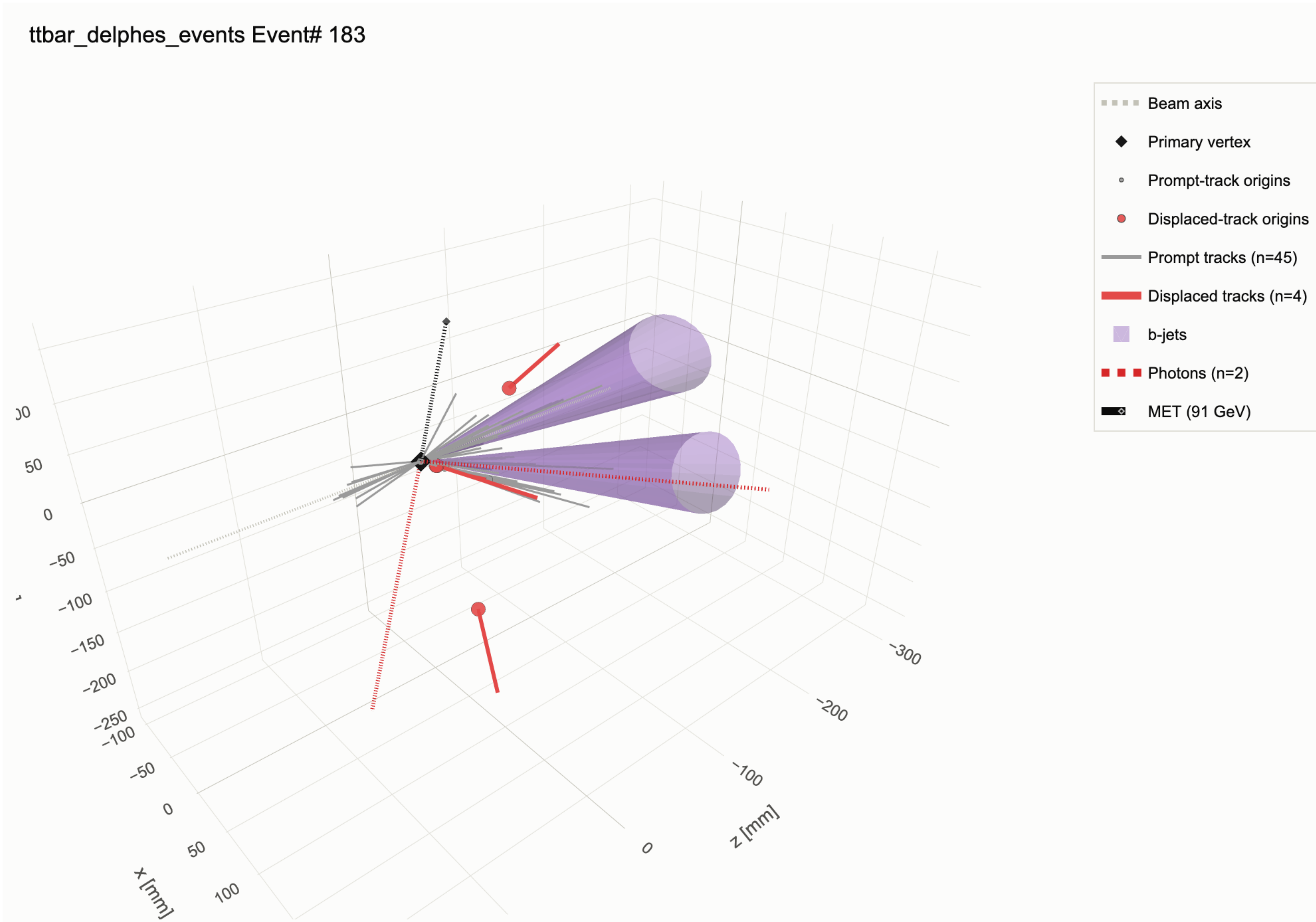
Tracks with |d0| above the threshold are coloured distinctly from prompt tracks.

Output: displaced tracks (polar view)



4 of 45 tracks flagged as displaced (|d0| > 1 mm), picked out in red

Output: displaced tracks (3D view)



Same event -- displaced-track origins marked distinctly in 3D

Vertex displays: loading a pileup ntuple

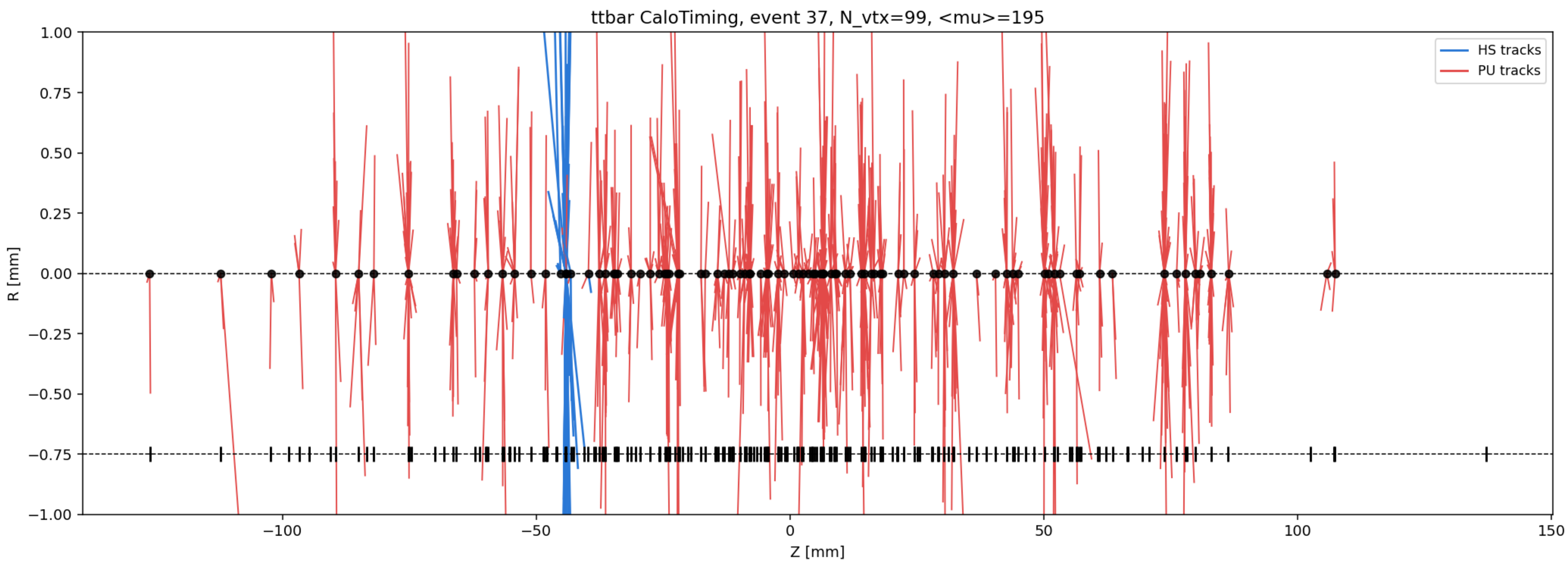
```
from tracehep.io.calotiming import load_vertex_event

vtx_event = load_vertex_event(
    "calo_timing_ntuple.root",
    event_index=37,
)
print(len(vtx_event.vertices), "vertices")
print(vtx_event.mu, "<mu>")
# -> 99 vertices
# -> 195.0 <mu>
```

The z-R vertex display, three styles

```
from tracehep import plot_vertices_zr

plot_vertices_zr(vtx_event, style="plain").savefig("plain.png")
plot_vertices_zr(vtx_event, style="styled").savefig("styled.png")
plot_vertices_zr(vtx_event, style="time_colored").savefig("time.png")
```

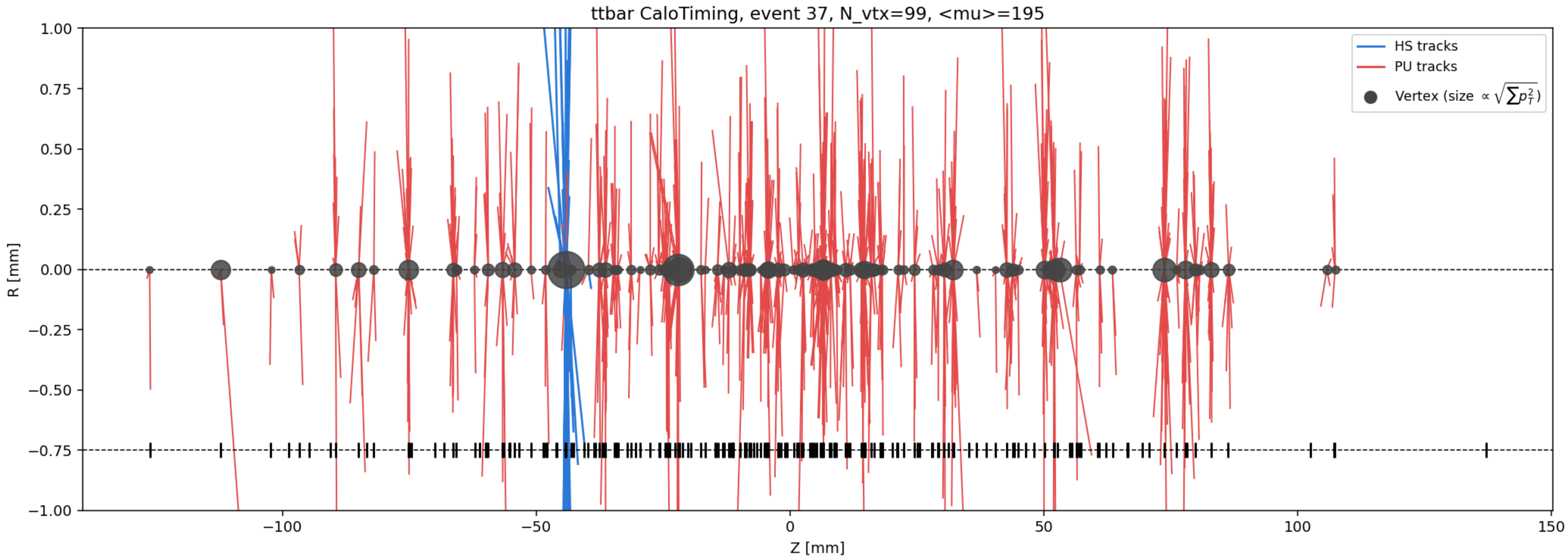
style is one of "plain", "styled", or "time_colored".

Output: style="plain"



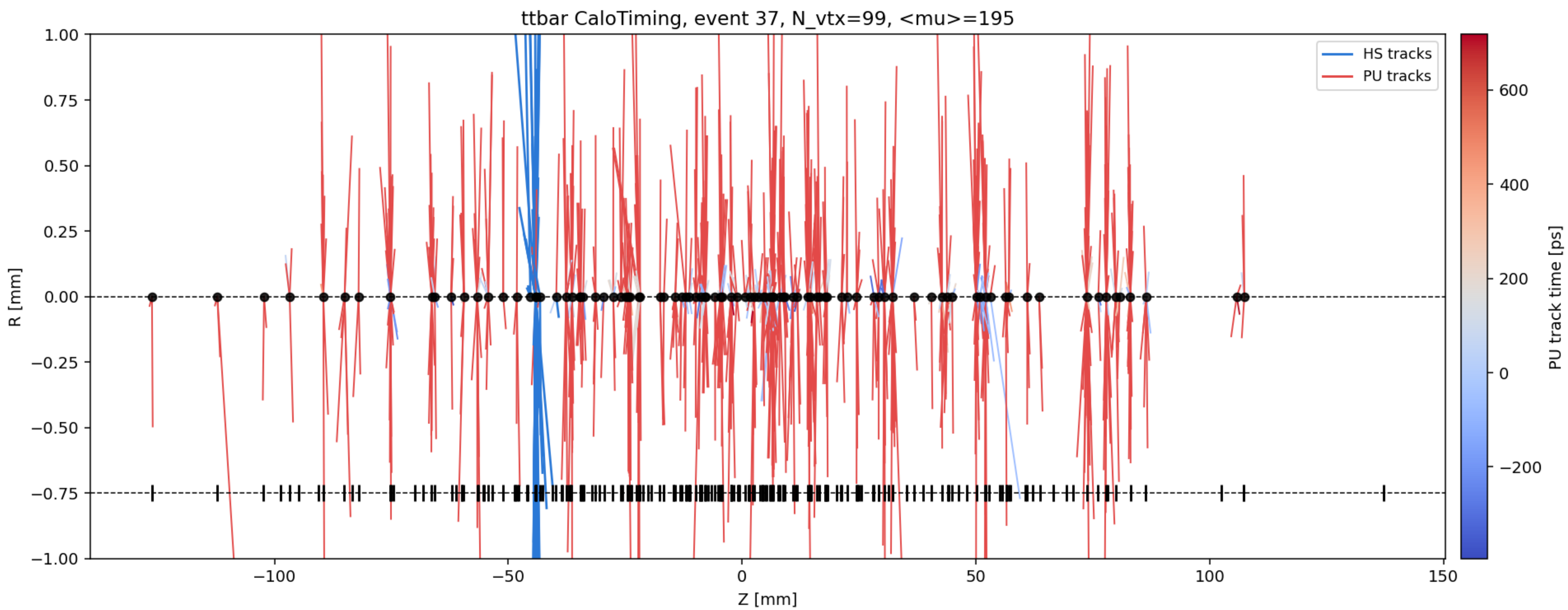
Flat HS (blue) / PU (red) track colouring, fixed vertex marker size

Output: style="styled"



Vertex marker size proportional to sqrt(sum pT^2); <mu> reported in the title

Output: style="time_colored"



PU tracks coloured by reconstructed track time (needs per-track timing)

Zooming into one vertex

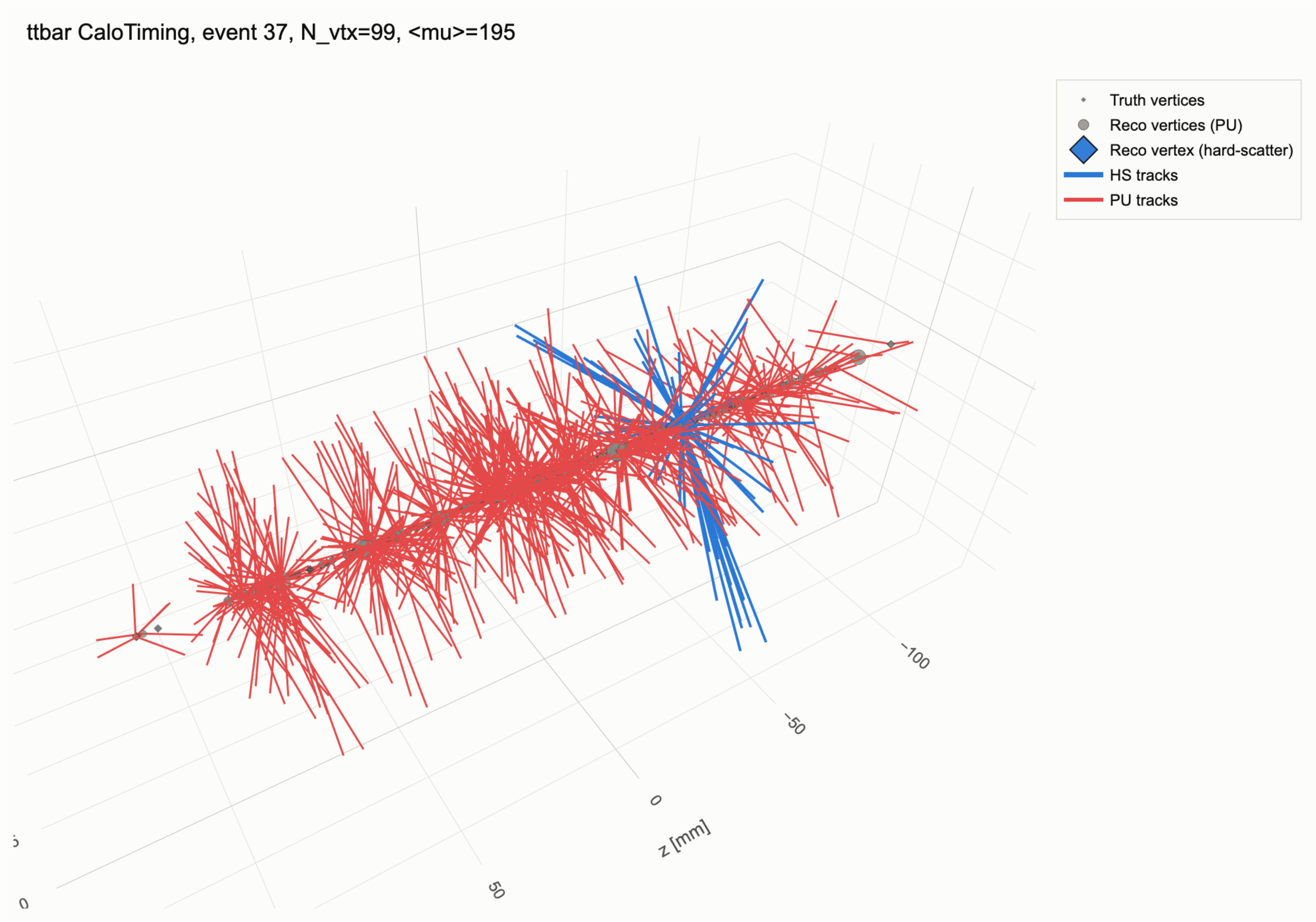
```
plot_vertices_zr(
    vtx_event, style="styled",
    zoom_center=-44.2, zoom_range_mm=5.0,
)
```

Same function -- pass zoom_center/zoom_range_mm to reproduce a single-vertex zoomed view.

Interactive 3D vertex display

```
from tracehep import plot_vertices_3d

fig = plot_vertices_3d(vtx_event)
fig.write_html("vertices_3d.html")
```

Output: the interactive 3D vertex view



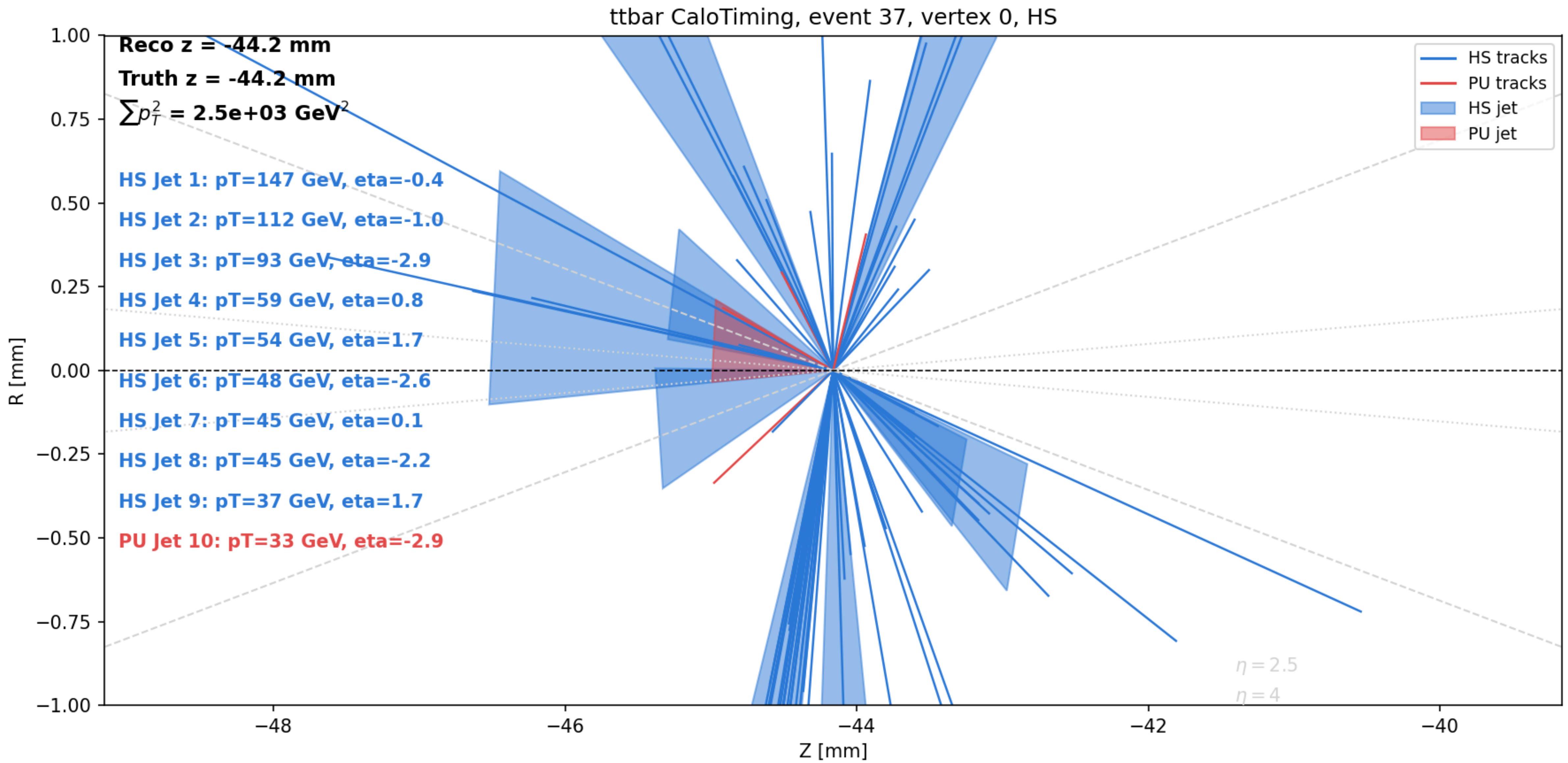
Every reconstructed and truth vertex at its real (x, y, z) position

One vertex, with its associated jets

```
from tracehep import plot_vertex_detail
from tracehep.io.calotiming import match_jets_to_vertex

vtx = vtx_event.vertices[0]
jets = match_jets_to_vertex(
    "calo_timing_ntuple.root", event_index=37, vtx_z=vtx.z,
)

fig = plot_vertex_detail(vtx_event, vtx_index=0, jets=jets)
fig.savefig("vertex0_detail.png")
```

match_jets_to_vertex associates jets by Rpt (track-pT-fraction) matching -- bring your own jets for a different scheme.

Output: single-vertex detail with jets



Vertex 0 (HS): 10 jets matched (9 truth-HS, 1 truth-PU), sum(pT^2) = 2.5e3 GeV^2 -- tracks and jets each coloured by their own truth match

Filtering and jet collections

Real analyses rarely want every jet or every track --
trace-hep supports both, without touching a single plotting function:

1) Jet collection is a LOADER choice
-- which ROOT branches to read (Jet, GenJet, JetPUPPI, ...)

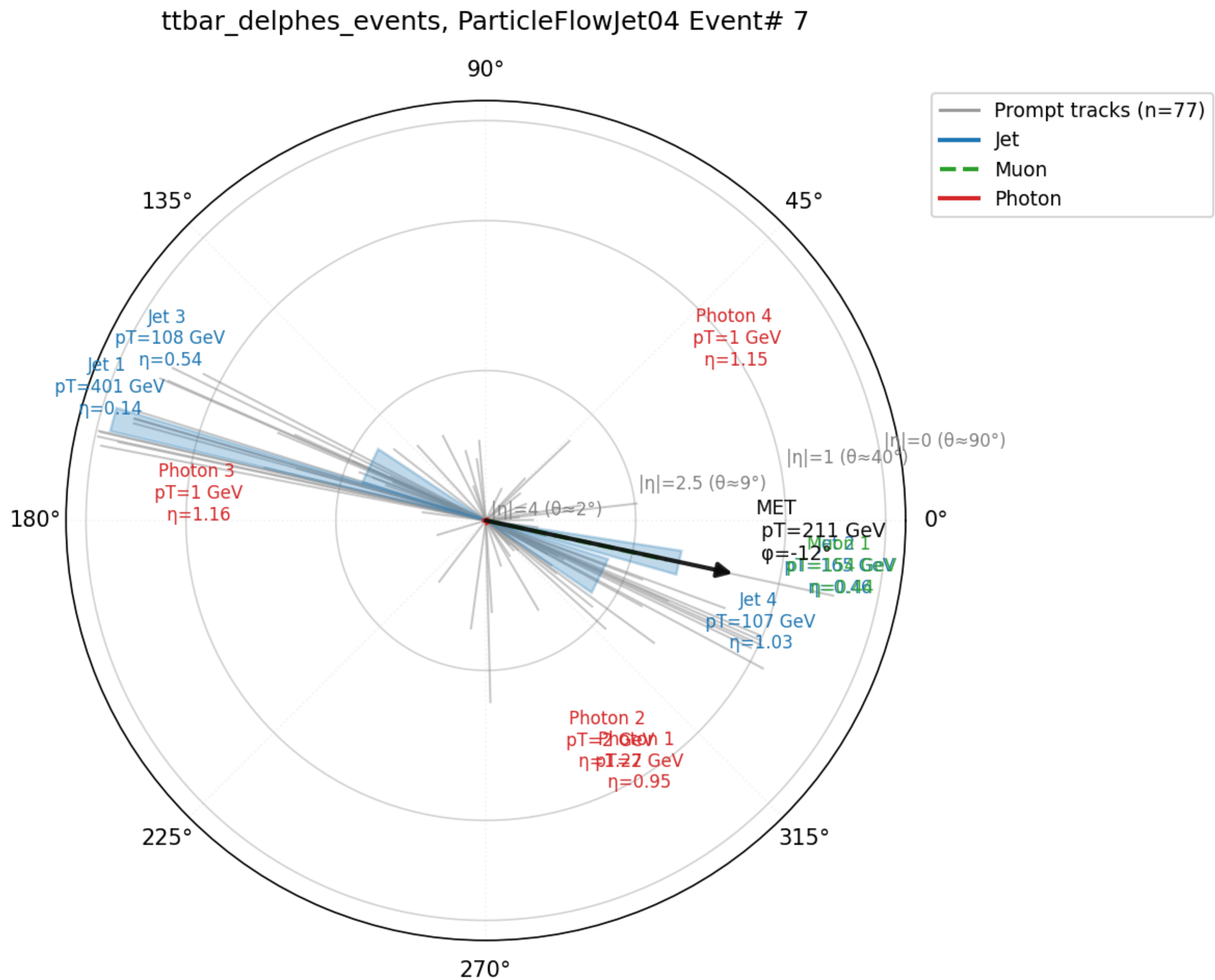
2) pT / eta cuts are a POST-LOAD, format-agnostic filter
-- tracehep.filters transforms an already-loaded Event or
VertexEvent, so one cut applies identically to every
display: polar, beam2d, 3D, z-R, vertex-detail, ...

Choosing a jet collection

```
from tracehep.io.delphes import load_events

# a Delphes card can define several jet collections --
# default is "Jet"; read a different one by name
events = load_events(
    "ttbar_delphes_events.root", indices=[7],
    jet_collection="ParticleFlowJet04",
)
trace.plot_event_polar(events[7], show_tracks=True)
```

Also works for match_jets_to_vertex (jet_collection="AntiKt4EMPFlowJets", ...) and the flat-ntuple loader (jet_branches=(...), bjet_branches=(...)).

Output: ParticleFlowJet04 vs. the default Jet collection



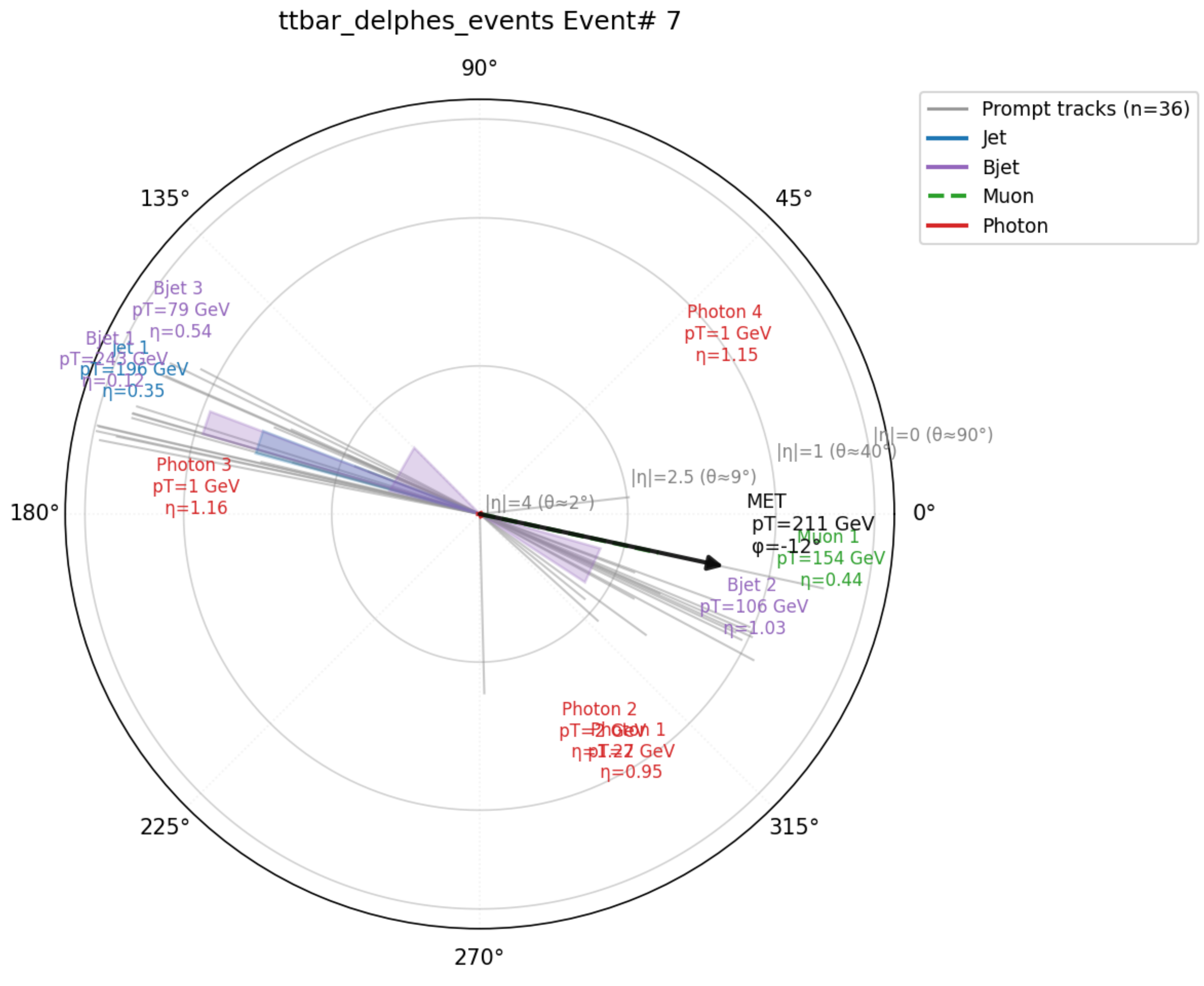
Same event, different jet collection: Jet 1 = 401 GeV here vs. 243 GeV with the default "Jet" collection -- a different clustering algorithm/input, same event

Cutting on pT and eta

```
from tracehep.filters import filter_event

event = load_events("ttbar_delphes_events.root", indices=[7])[7]

# keep only jets above 50 GeV within |eta| < 2.5,
# and tracks above 2 GeV within |eta| < 2.5
tight = filter_event(
    event,
    jet_pt_min=50.0, jet_eta_min=-2.5, jet_eta_max=2.5,
    track_pt_min=2.0, track_eta_min=-2.5, track_eta_max=2.5,
)
trace.plot_event_polar(tight, show_tracks=True)
```

eta_min/eta_max bound eta directly (signed, not abs(eta)) -- pass eta_min=0 to keep only positive-eta objects.

Output: after pT/eta cuts



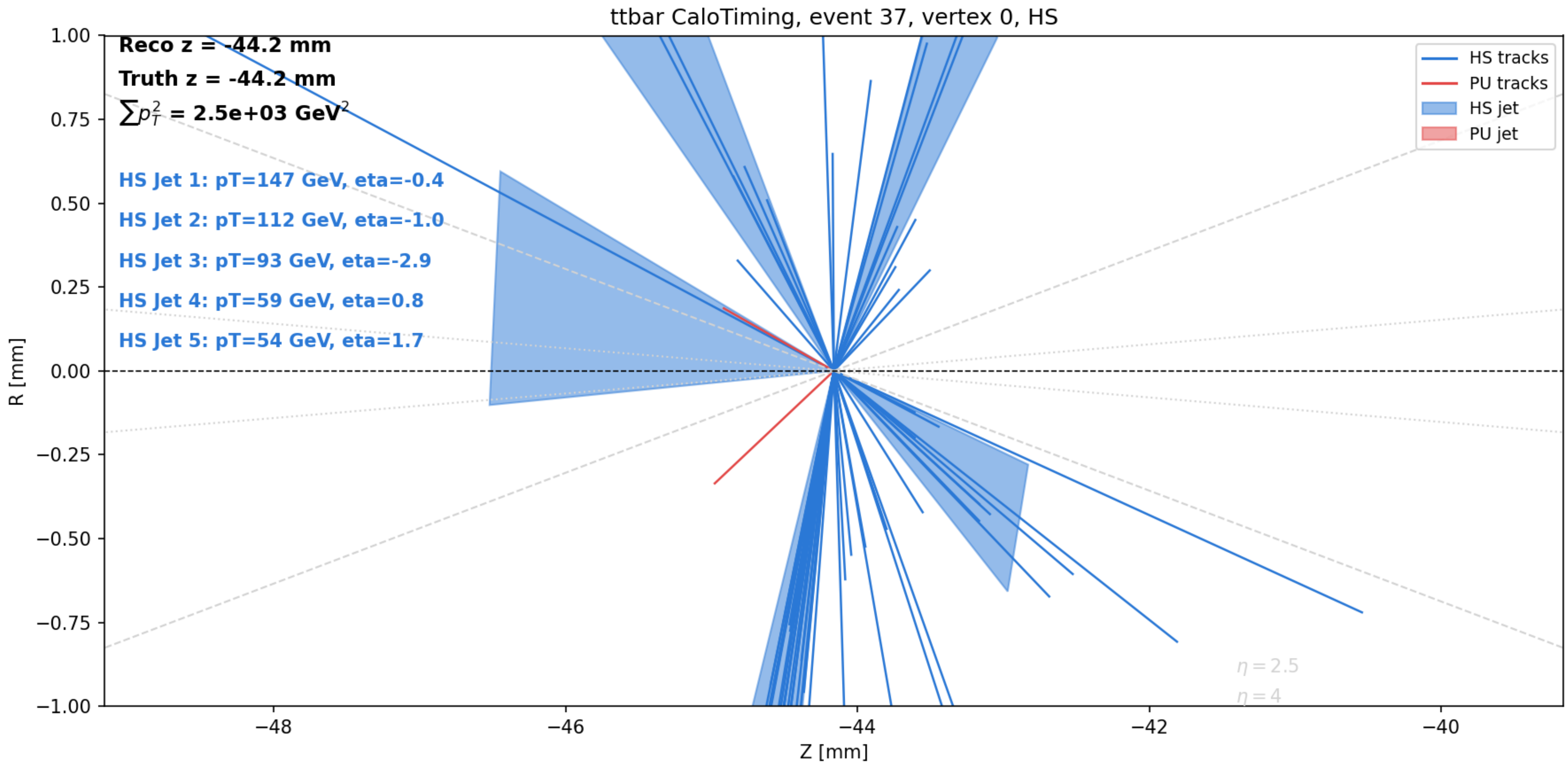
Same event 7: 77 -> 36 tracks after the pT/eta cut (all 4 jets already passed pT>50 GeV, |eta|<2.5)

Filtering a vertex scenario

```
from tracehep.filters import filter_vertex_event

vtx_event = load_vertex_event("calo_timing_ntuple.root", event_index=37)
vtx_event = filter_vertex_event(
    vtx_event, track_pt_min=1.0, track_eta_min=-2.5, track_eta_max=2.5,
)
plot_vertices_zr(vtx_event, style="styled")
```

Vertices are never dropped, only the tracks fit to them -- each vertex's track_indices is remapped to stay consistent.

Output: vertex 0 after track pT/eta cuts



Event 37: 969 -> 607 tracks event-wide after the cut; vertex 0 goes from 10 to 5 matched jets once jet_pt_min=50 GeV is also applied

Filtering from the command line

```
trace-batch --input sample.root --indices 0 1 2 --outdir out/ \
    --jet-collection ParticleFlowJet04 \
    --jet-pt-min 30 --jet-eta-min -2.5 --jet-eta-max 2.5 \
    --show-tracks --track-pt-min 1.0 \
    --track-eta-min -2.5 --track-eta-max 2.5
```

Every filter available in Python is also a trace-batch flag --
no code needed for a batch of cut, collection-specific displays.

Real public data: ATLAS Open Data

tracehep.io.atlas_opendata reads the public ATLAS Open Data 13 TeV "mini-ntuple" format (opendata.atlas.cern) directly -- no Delphes, no private ntuple, no ROOT/CMSSW build needed.

The same flat schema is used for EVERY released 13 TeV dataset -- SM, Higgs, SUSY, exotic-resonance searches, ... -- so one loader works unchanged across all of them.

Find file URLs with the companion atlasopenmagic package rather than guessing paths by hand: pip install "trace-hep[opendata]"

Finding a dataset URL

```
import atlasopenmagic as atom

atom.set_release("2020e-13tev")
atom.available_keywords()      # -> ttbar, Higgs, Zprime, susy, ...

# ttbar, dilepton skim
urls = atom.get_urls("410000", skim="2lep", protocol="https")
```

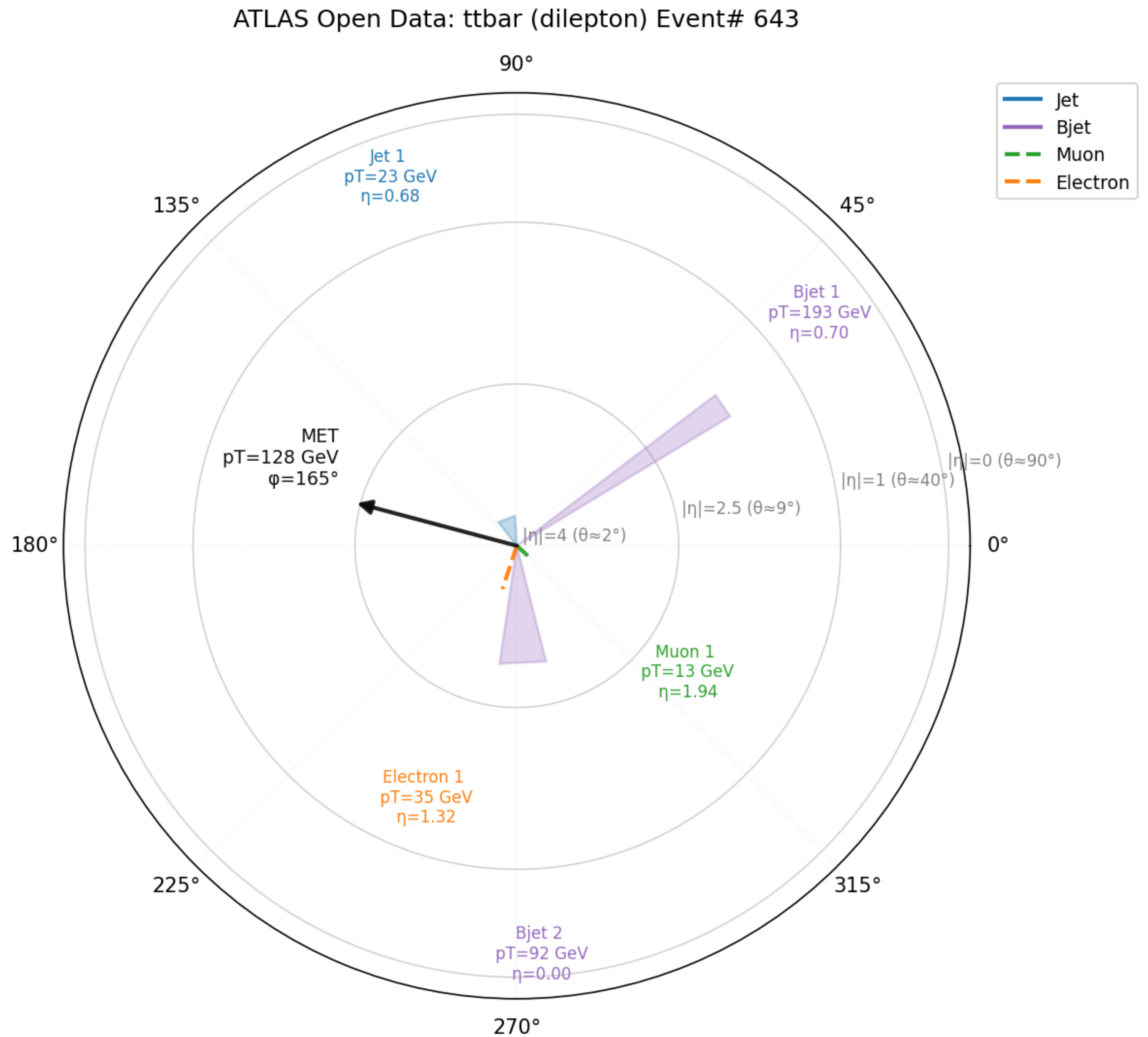
atlasopenmagic also covers 2024/2025 research-grade releases, not just the 2020 education set used here.

Loading and drawing an ATLAS Open Data event

```
from tracehep.io.atlas_opendata import load_events

events = load_events(
"mc_410000.ttbar_lep.2lep.root", indices=[0, 1],
label="ttbar (ATLAS Open Data)",
)
trace.plot_event_polar(events[1]).savefig("event1_polar.png")
```

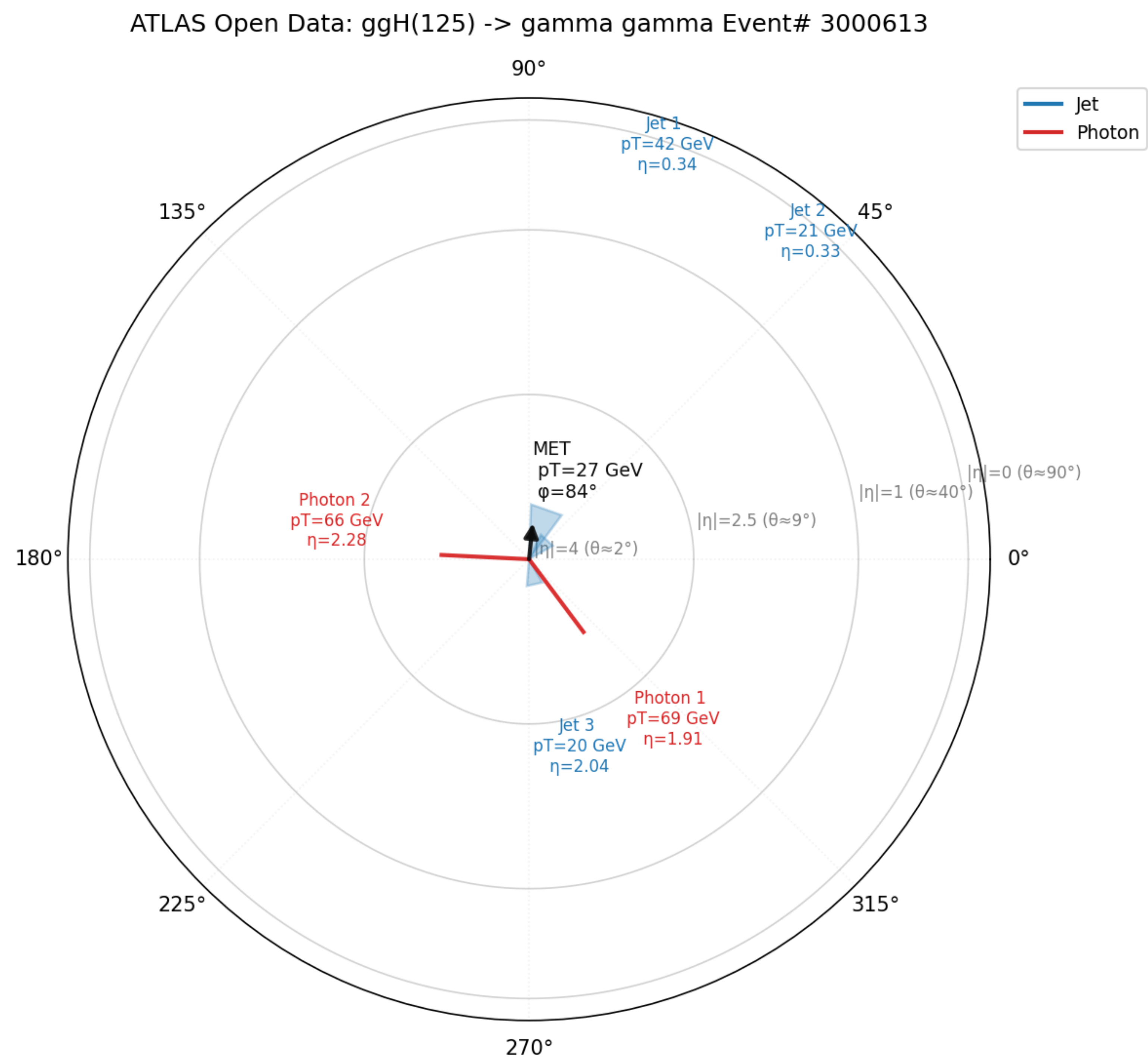
Momenta are stored in MeV in this ntuple -- the loader converts to GeV automatically.

Output: a real ttbar (e-mu) event



Genuine ATLAS Open Data dilepton ttbar candidate: 2 b-jets, one electron, one muon, and real MET -- no simulation shortcuts

Output: a real H -> gamma-gamma candidate



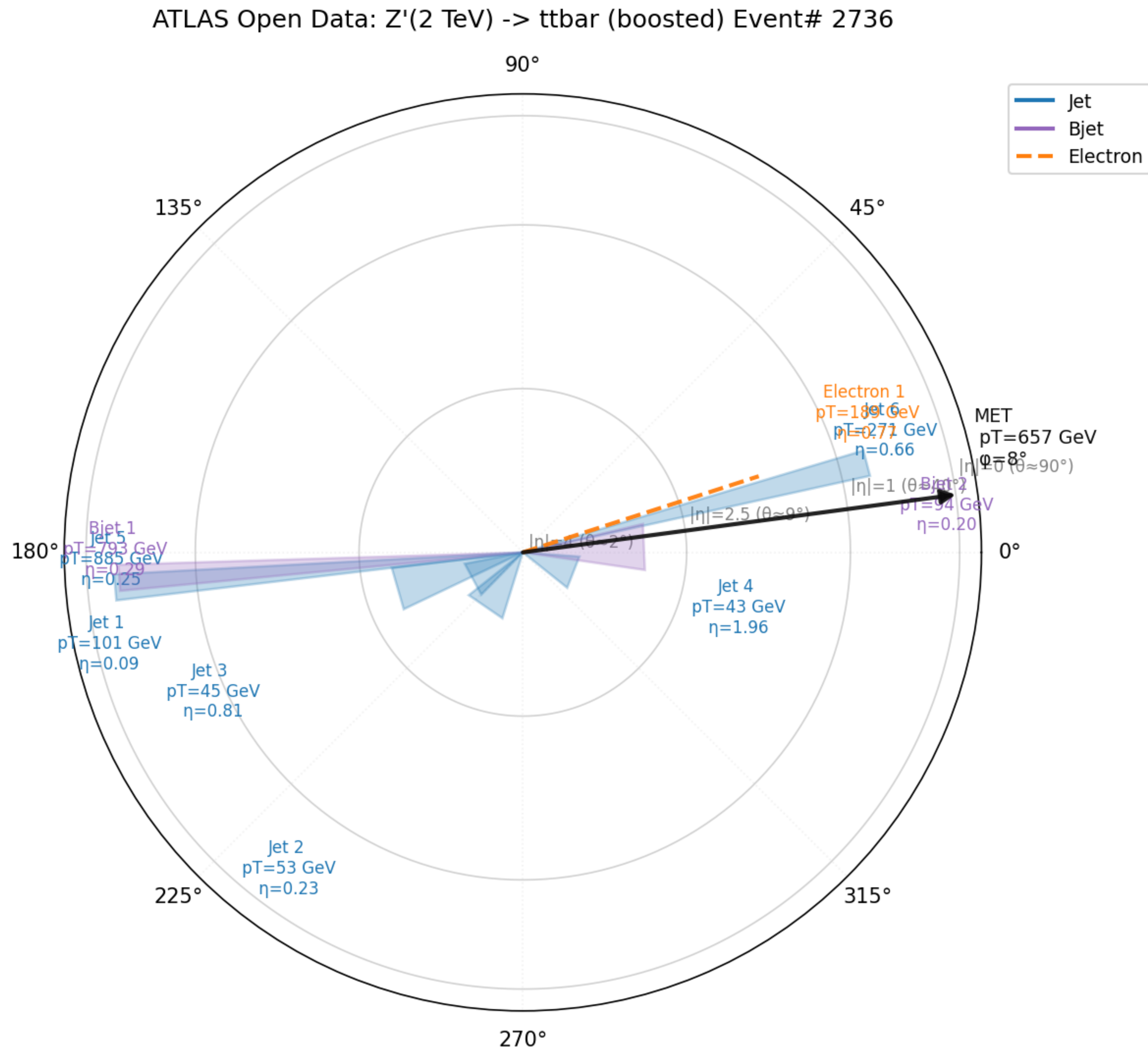
ggH(125) -> gamma-gamma skim: exactly two photons per event, plus the underlying-event jets

Boosted topologies: large-R jets

```
events = load_events(
"mc_301329.ZPrime2000_tt.1largeRjet1lep.root", indices=[4],
include_large_r_jets=True,
label="Z'(2 TeV) -> ttbar (boosted)",
)
trace.plot_event_polar(events[4])
```

include_large_r_jets=True appends the largeRjet_* collection -- b-tagging (MV2c10) is only defined for small-R jets.

Output: a boosted Z' -> ttbar event



2 TeV resonance, 657 GeV MET -- the large-R jet label overlaps its small-R constituents, as expected: they point the same direction by construction

Real public data: CMS Open Data

tracehep.io.cms_opendata reads the public CMS Open Data reduced "NanoAOD outreach" format (opendata.cern.ch) -- 2012, 8 TeV.

Covers both MC samples (TTbar, SMHiggsToZZTo4L, ZZTo4mu, ...) AND REAL 2012 LHC collision data (Run2012B_DoubleMuParked, ...).

Different sub-releases carry different object collections (some have Jet+Muon+Tau, some have Muon+Electron only) -- the loader detects what's actually present rather than requiring per-sample configuration.

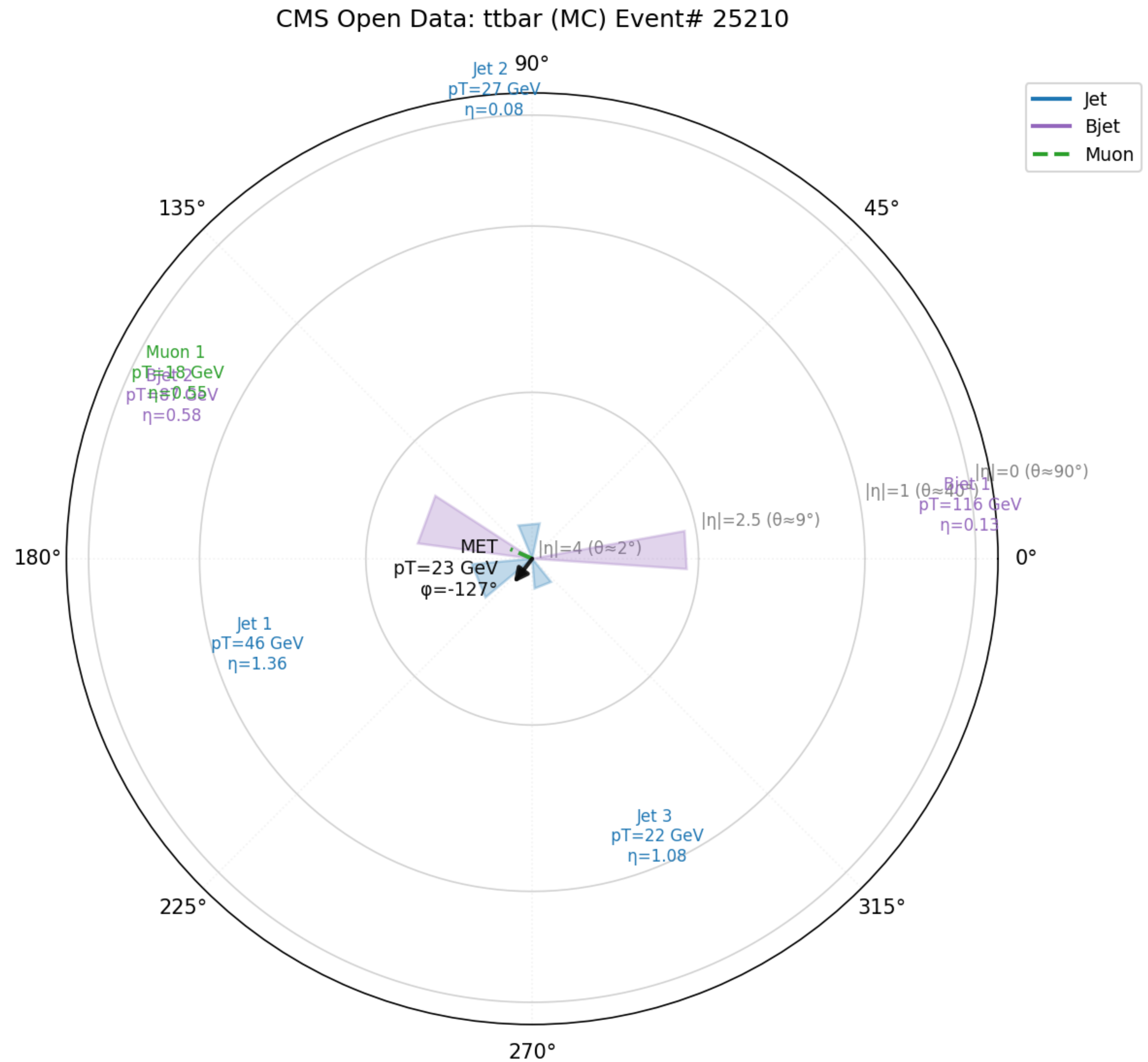
Large files stream directly over HTTPS -- uproot reads only the bytes it needs, no full download required.

Streaming a multi-GB file with no download

```
from tracehep.io.cms_opendata import load_events

url = ("https://opendata.cern.ch/eos/opendata/cms/"
       "derived-data/AOD2NanoAODOutreachTool/TTbar.root")

events = load_events(url, indices=[9], label="ttbar (CMS Open Data)")
trace.plot_event_polar(events[9])
```

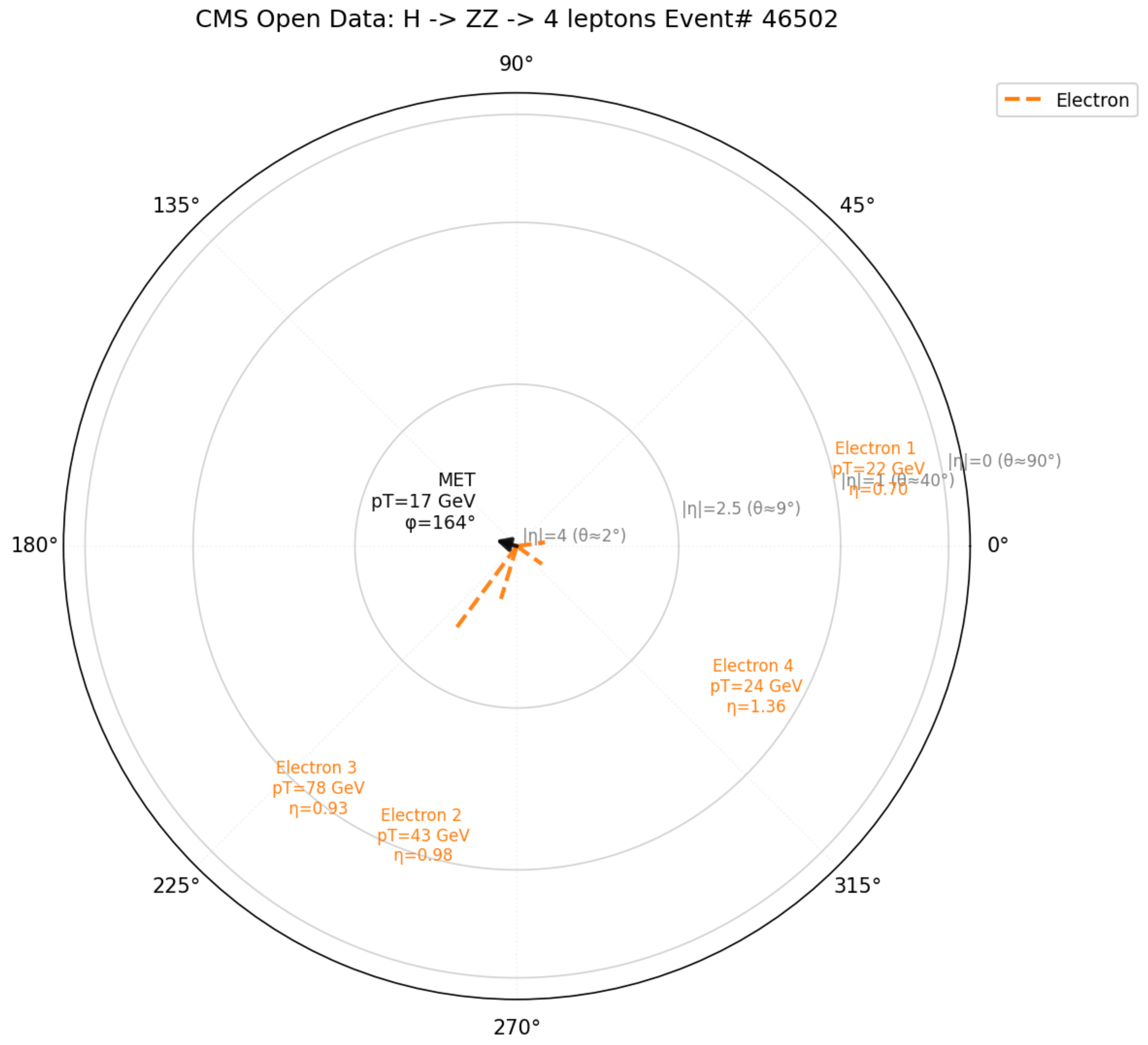
TTbar.root is 3.5 GB on disk -- uproot streams only the baskets needed for these 3 events.

Output: a real ttbar-like event (streamed)



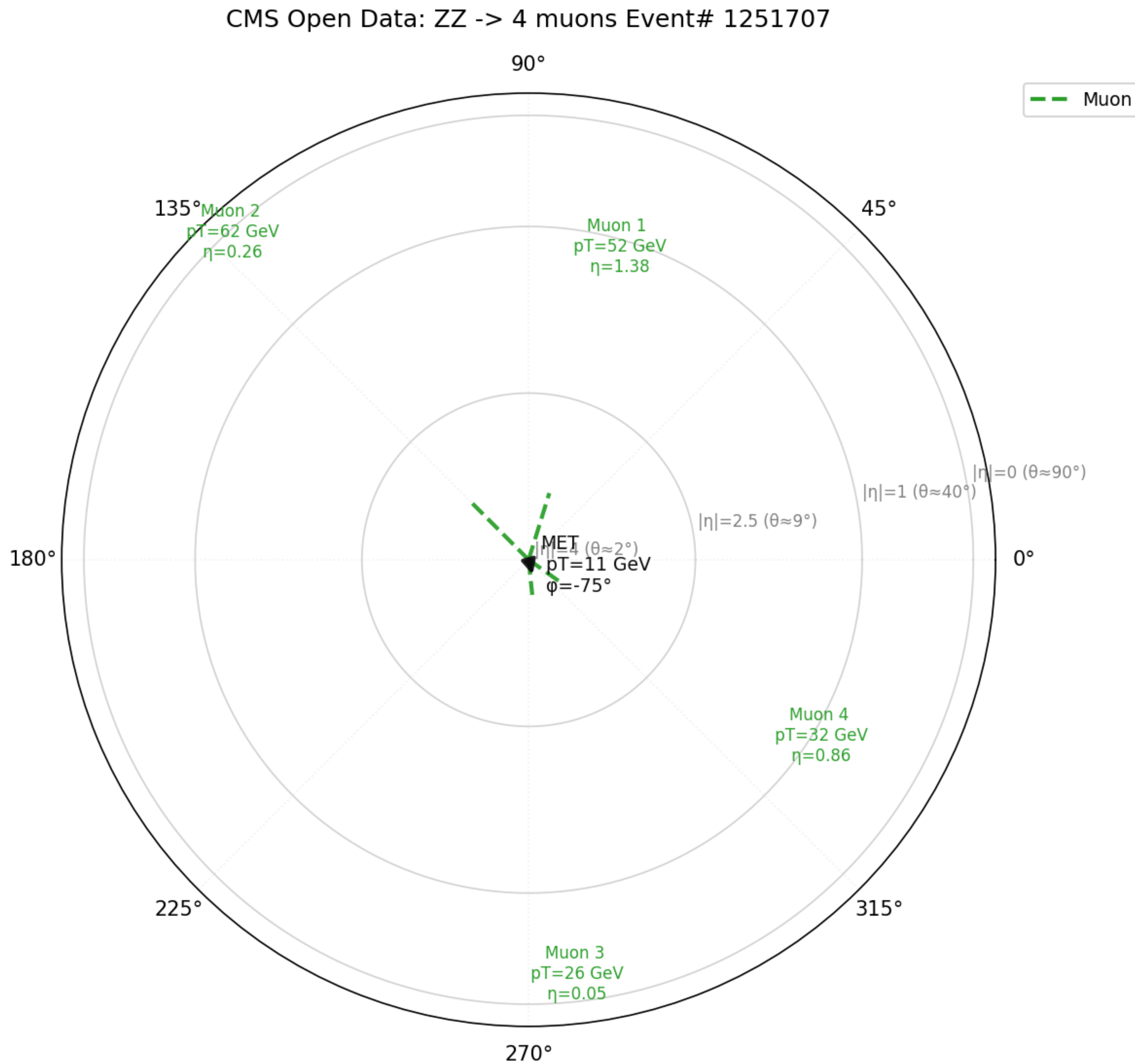
2 b-tagged jets, 1 muon, MET -- read directly from opendata.cern.ch, never downloaded to disk

Output: H -> ZZ -> 4 leptons candidate



SMHiggsToZZTo4L sample: a clean 4-electron candidate

Output: ZZ -> 4 muons candidate



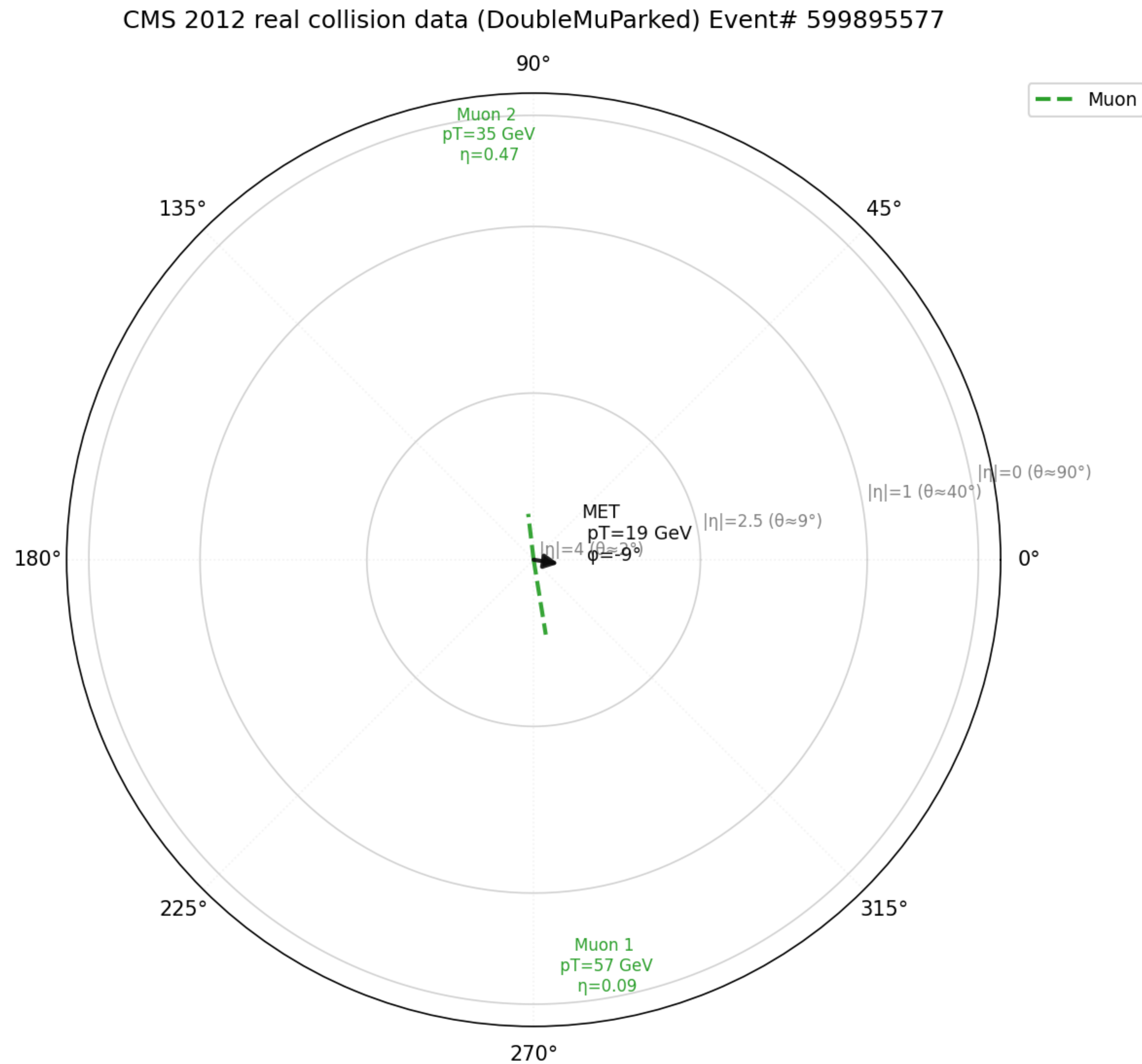
ZZTo4mu sample: four muons, a textbook diboson topology

Real collision data, not simulation

```
url = ("https://opendata.cern.ch/eos/opendata/cms/derived-data/"
       "AOD2NanoAODOutreachTool/ForHiggsTo4Leptons/"
       "Run2012B_DoubleMuParked.root")

events = load_events(url, indices=[2], label="CMS 2012 real data")
print(events[2].run, events[2].event_number)  # -> 194108 599895577
```

A real LHC run/event number, not a MC placeholder -- this is an actual proton-proton collision recorded by CMS in 2012.

Output: a genuine 2012 LHC collision event



Run 194108, event 599895577 -- two real, reconstructed muons from an actual CMS-recorded collision, not Monte Carlo

Can we get real public vertex/pileup displays?

Checked directly against 200 real events before building anything:

ATLAS/CMS education releases: no track or vertex branches at all.

ATLAS's 2024 PHYSLITE research release DOES have InDetTrackParticles (d0, z0, ...) and PrimaryVertices (x, y, z, vertexType) -- but PHYSLITE thins away most tracks not tied to a lepton/jet. Measured: 199/200 hard-scatter vertices keep real tracks; of 2664 pileup vertices, only ~3.6% retain any tracks at all (1-12 each).

CMS full AOD has complete track/vertex collections, but reading it needs the CMSSW software framework, not just uproot.

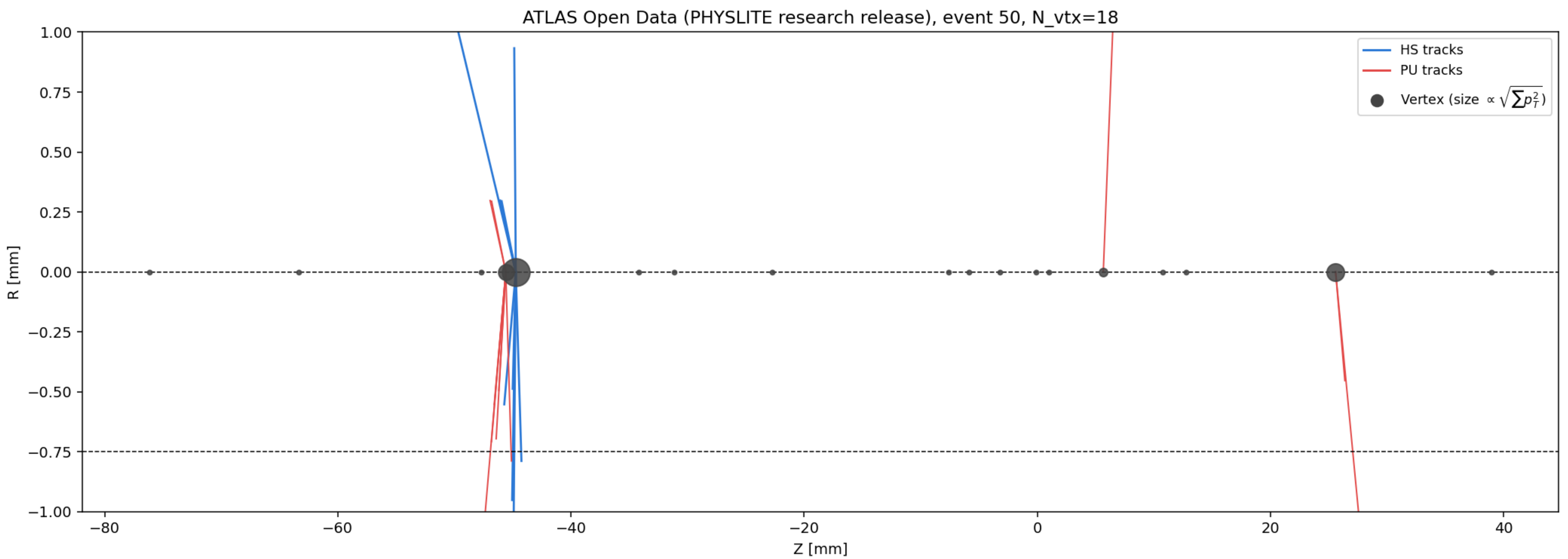
Conclusion: a real single-vertex detail plot IS possible (below), and a many-vertex survey occasionally shows a pileup vertex with real tracks too -- but nowhere near as dense as the mu~195 private-ntuple survey, which has no public open-data equivalent at that density.

A real public vertex display (ATLAS PHYSLITE)

```
from tracehep.io.atlas_physlite import load_vertex_event

url = ("https://opendata.cern.ch/eos/opendata/atlas/rucio/"
       "mc20_13TeV/DAOD_PHYSLITE.37620644._000012.pool.root.1")
vtx_event = load_vertex_event(url, event_index=50)

trace.plot_vertices_zr(vtx_event, style="styled")
```

Event 50 chosen because it's one of the ~3.6% of events with real pileup-vertex tracks -- most won't show any.

Output: every vertex's real position



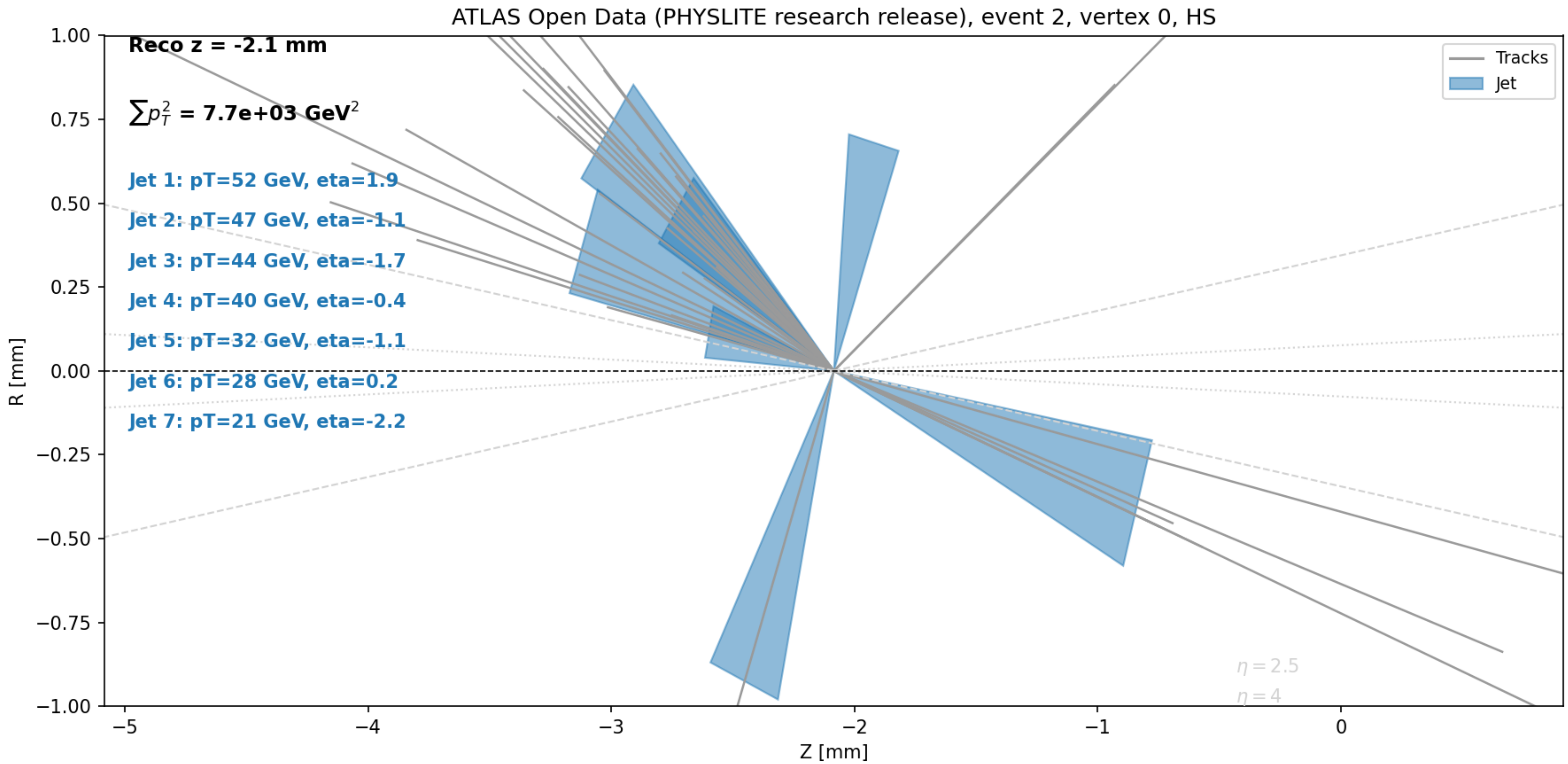
18 vertices, real (x,y,z) -- the hard-scatter vertex (centre) has a full track fan, and here 3 pileup vertices genuinely retained a few tracks too (red), unusually for this dataset

The hard-scatter vertex's real tracks and jets

```
from tracehep.io.atlas_physlite import load_event_jets

# a different event, picked for a richer hard-scatter vertex (38 tracks)
vtx_event = load_vertex_event(url, event_index=2)
hs_idx = next(i for i, v in enumerate(vtx_event.vertices) if v.is_hs)
jets = load_event_jets(url, event_index=2)  # not vertex-associated, no truth

trace.plot_vertex_detail(vtx_event, vtx_index=hs_idx, jets=jets, zoom_range_mm=3.0)
```

Neither tracks nor jets have is_hs set here -- plot_vertex_detail draws them in one flat colour each ("Tracks"/"Jet"), not a fabricated HS/PU split.

Output: genuine ATLAS-reconstructed tracks and jets



38 real tracks + 7 real jets, sum(pT^2) = 7.7e3 GeV^2 -- an actual ATLAS hard-scatter vertex from public research-grade data, not simulation; no truth here, so no HS/PU colour split is claimed

Batch processing many events at once

```
from tracehep.io.delphes import load_events
from tracehep.batch import run_event_batch

events = load_events(path, indices=range(20))
run_event_batch(
    events, "out/", which=("polar", "beam2d"),
    polar_kwargs={"show_tracks": True},
)
# -> out/event0_polar.png, out/event0_beam2d.png, ...
```

Batch processing vertex scenarios

```
from tracehep.io.calotiming import load_vertex_event
from tracehep.batch import run_vertex_batch

vtx_events = {i: load_vertex_event(path, i) for i in range(10)}
run_vertex_batch(vtx_events, "out/", styles=("plain", "styled"))
# -> out/vertices0_plain.png, out/vertices0_styled.png, ...
```

The trace-batch command-line tool

```
trace-batch \
--input ttbar_delphes_events.root \
--indices 0 1 2 \
--outdir out/ \
--which polar beam2d \
--show-tracks \
--label my_sample
```

No Python required -- useful for quick batch regeneration from a shell script.

Failure-mode & anomaly review

The recurring question: "algorithm 1 failed these events but algorithm 2 passed them -- what do they actually look like?" (or, for anomaly detection: "400 events flagged anomalous, let's eyeball them"). tracehep.gallery turns a list of event numbers + a category per event into ONE self-contained, portable HTML file.

No new dependency -- every image is embedded as base64 PNG directly in the HTML. No folder of hundreds of loose image files to manage; nothing is written to disk until you click Download.

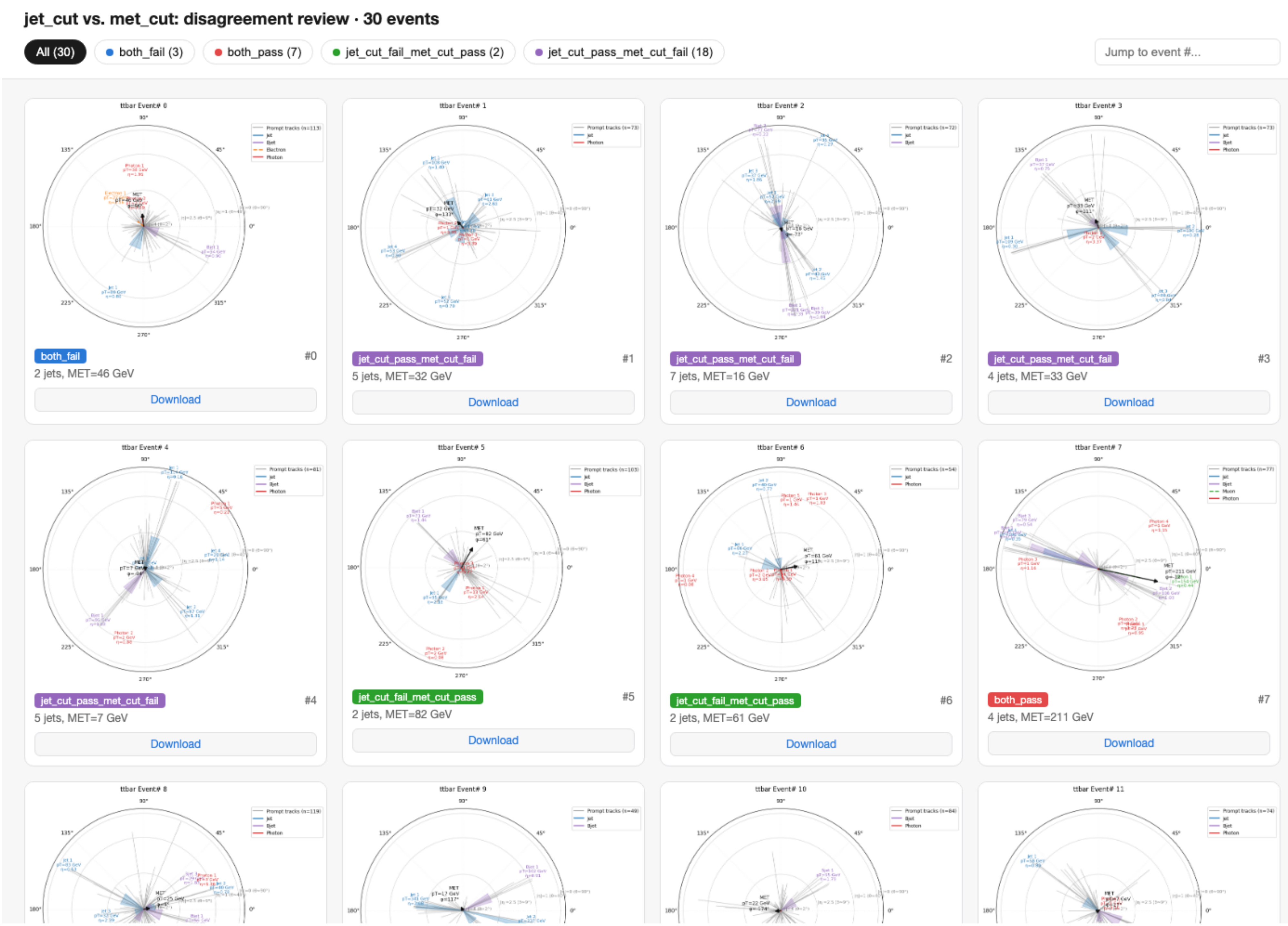
Open the file in any browser: filter by category, search/jump to an event number, click a thumbnail for a full-size lightbox you page through with the arrow keys, click Download on any image you want.

Building a review gallery

```
from tracehep.gallery import build_gallery, compare_pass_fail

events = load_events("my_sample.root", indices=range(500))

# your real per-event pass/fail results, e.g. two ML classifiers
algo1_pass = {i: my_classifier_1.passes(i) for i in events}
algo2_pass = {i: my_classifier_2.passes(i) for i in events}
categories = compare_pass_fail(algo1_pass, algo2_pass, name_a="clf1",
name_b="clf2")

build_gallery(
events, categories,
plot_fn=lambda ev: trace.plot_event_polar(ev, show_tracks=True),
output_path="review.html", title="clf1 vs. clf2 disagreement review",
)
```

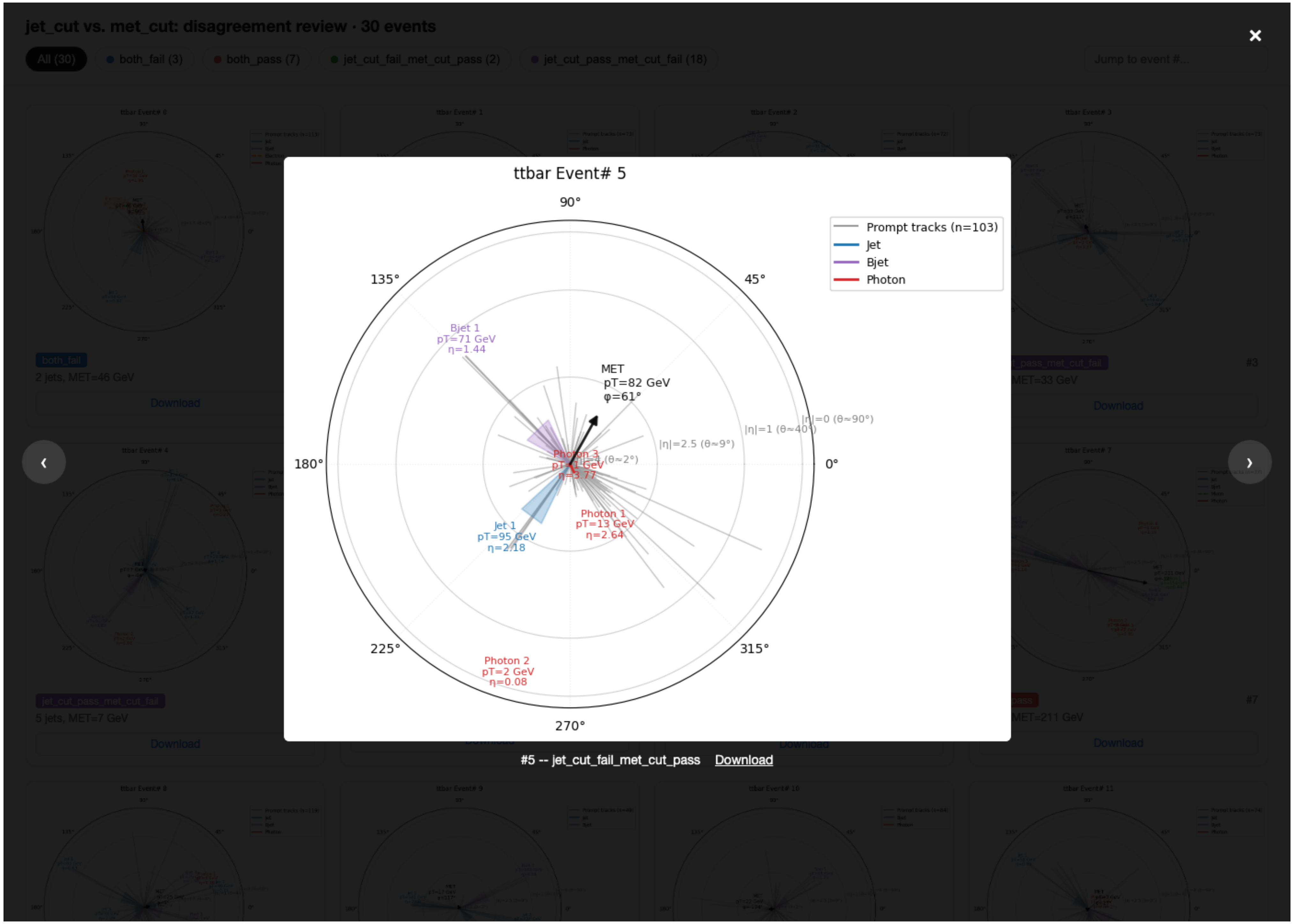
compare_pass_fail builds the 4-category split (both_pass/both_fail/clf1_pass_clf2_fail/clf1_fail_clf2_pass); for anomaly detection just pass {eid: "anomalous", ...} yourself.

Output: the browsable gallery



30 events, categorized by two toy "algorithms" (a jet-count cut vs. a MET cut) -- filter tabs with live counts, per-event captions, click-to-download

Output: the full-size lightbox



Click any thumbnail to page through full-size with the arrow keys -- exactly the "scroll through many events" review workflow, without saving hundreds of files

Live interactive viewer: trace-gui

For "just let me look at one event" -- no script, no pre-rendering. trace-gui starts a small local web app in your browser: type a file path (or URL) and an event number, pick a loader and a display, click Render, it draws live.

Covers every loader tracehep ships -- Delphes, flat-ntuple, calo-timing, ATLAS Open Data, CMS Open Data, ATLAS PHYSLITE -- and both 2D (matplotlib) and interactive 3D (plotly, real WebGL) displays.

Every render is a shareable URL (?path=...&loader=...&event_index=...) -- paste a link straight to one event instead of re-typing the form.

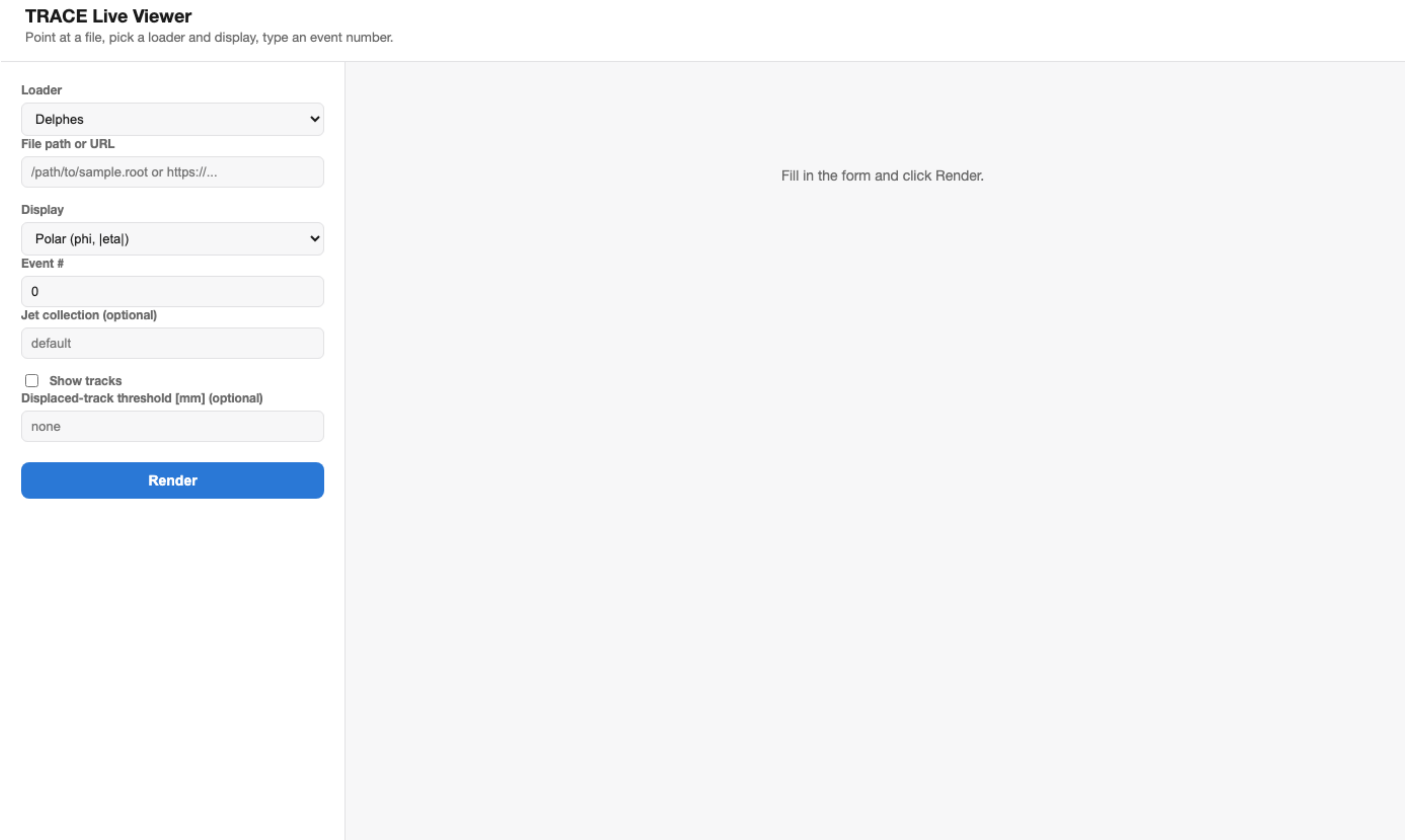
One new dependency: Flask, via pip install "trace-hep[gui]".

Launching it

```
pip install "trace-hep[gui]"
trace-gui

# -> TRACE viewer running at http://127.0.0.1:5057/
#    (opens automatically in your browser)
```

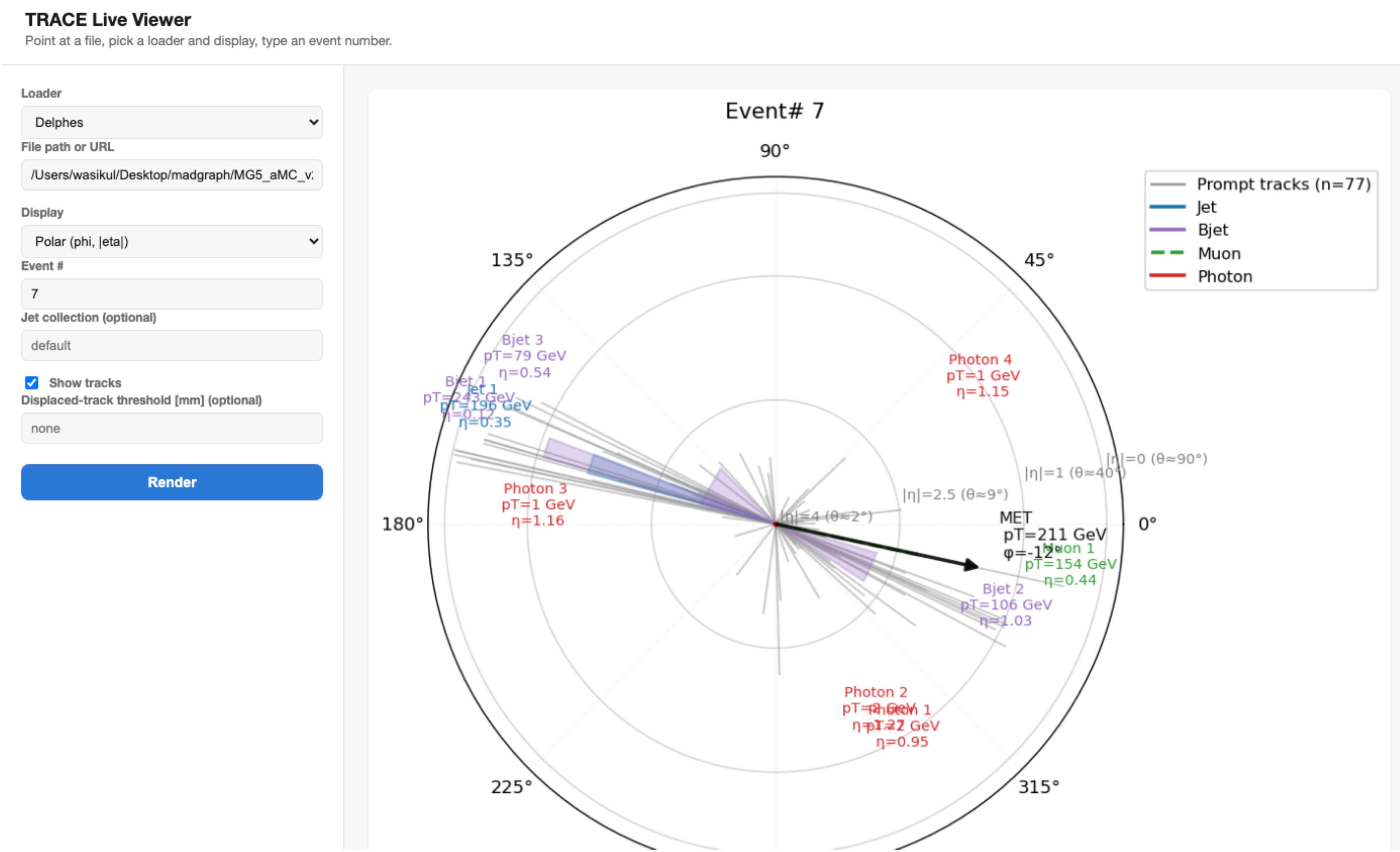
--port/--host/--no-browser flags available; run python3 -m tracehep.webapp as an alternative to the console script.

Output: the live viewer



Pick a loader, the Display dropdown updates to that format's valid displays automatically

Output: a real event, rendered live



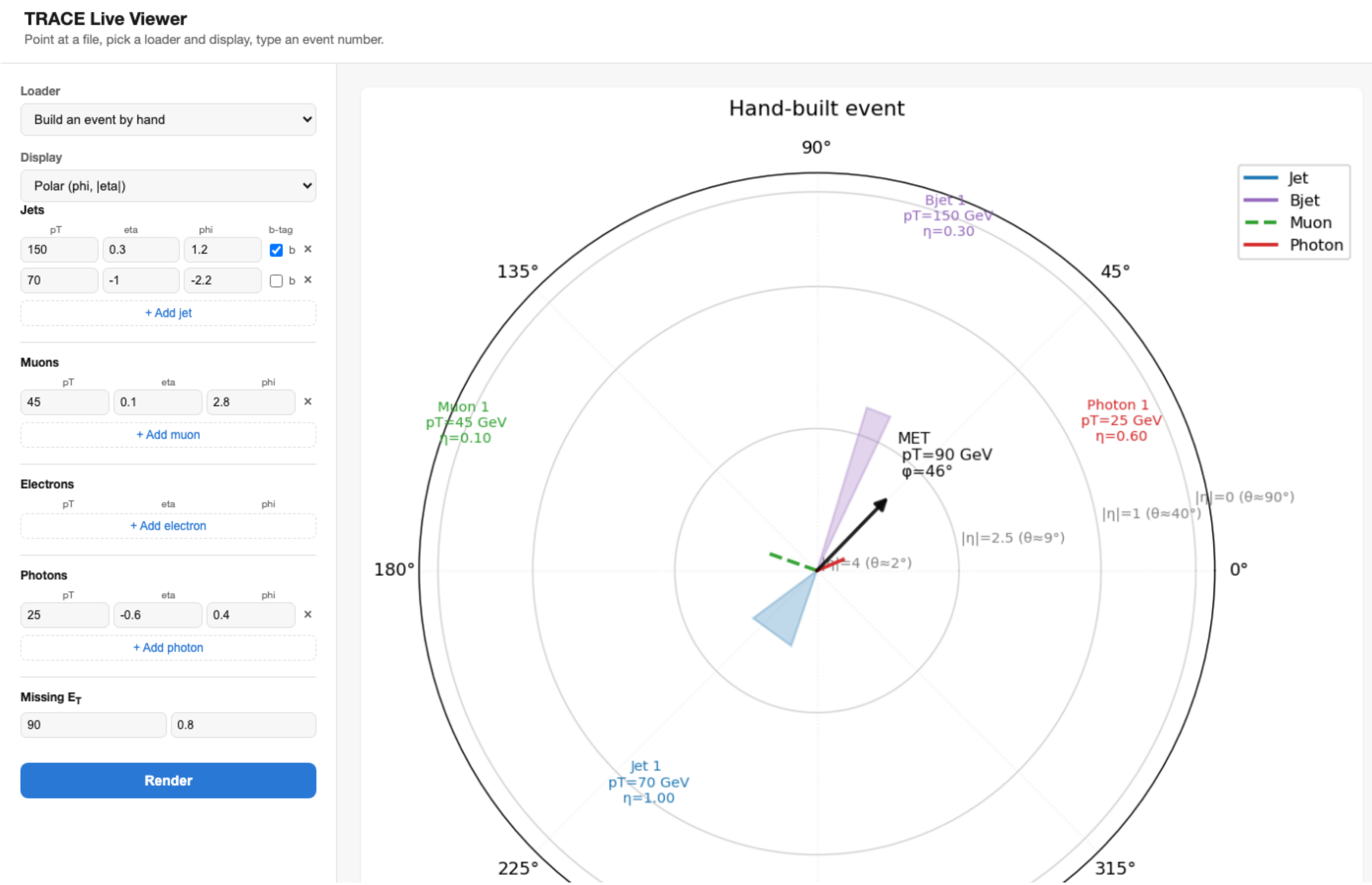
Same ttbar Delphes event #7 seen earlier in this deck -- loaded and drawn on demand, not pre-rendered

Build an event by hand -- no file needed

A dedicated loader mode: skip files entirely. Add jets, muons, electrons, photons, and MET by typing their pT/eta/phi directly in the browser, then render the same polar/beam2d/3D displays from exactly what you typed.

Useful for teaching ("here's what a boosted dijet topology looks like"), or for sanity-checking a hypothetical event shape without ever touching a ROOT file.

Output: a hand-built event, rendered live



2 jets (one b-tagged), 1 muon, 1 photon, MET -- typed directly into the form, no file, no script

Function reference: event displays

```
plot_event_polar(event, *, ax=None, eta_max=4.0, pt_scale=0.003,
        eta_guides=(0,1,2.5,4.0), show_tracks=False,
    track_pt_scale=0.15, d0_displaced_mm=None, title=None)

        plot_event_beam2d(event, *, ax=None,
    eta_guides=(2.5,4.0), angle_scale=4.0, title=None)

    plot_event_3d(event, *, d0_displaced_mm=None,
        jet_half_angle_deg=9.0, title=None)
```

Function reference: vertex displays

```
plot_vertices_zr(vertex_event, *, style="plain",
      zoom_center=None, zoom_range_mm=None,
             ax=None, title=None)

plot_vertex_detail(vertex_event, vtx_index, jets=(),
      *, zoom_range_mm=5.0, ax=None, title=None)

plot_vertices_3d(vertex_event, *, title=None)

run_event_batch(events, outdir, *, which=(...),
      polar_kwargs=None, beam2d_kwargs=None)
run_vertex_batch(vertex_events, outdir, *, styles=(...))
```

Loader reference (tracehep.io) 1/4

```
delphes.load_event(path, index, with_tracks=True,
            jet_collection="Jet", label="")
delphes.load_events(path, indices, with_tracks=True,
            jet_collection="Jet", label="")

flat_ntuple.load_event_by_run_event(path, run, event,
tree_name="Ntuple", jet_branches=("JET_pt","JET_eta","JET_phi"),
bjet_branches=("bJET_pt","bJET_eta","bJET_phi"), label="")
flat_ntuple.load_events_by_run_event(path, pairs, ...)

calotiming.load_vertex_event(path, event_index,
            tree_name="ntuple", label="")
calotiming.match_jets_to_vertex(path, event_index, vtx_z,
*, jet_collection="AntiKt4EMTopoJets", track_idx_branch=None,
        jet_pt_min=30.0, rpt_min=0.02, sig_cut=3.0)

All four require: pip install "trace-hep[delphes]"
```

Loader reference (tracehep.io) 2/4

```
atlas_opendata.load_event(path, index, tree_name="mini",
    btag_cut=0.6459, include_large_r_jets=False, label="")
atlas_opendata.load_events(path, indices, tree_name="mini",
    btag_cut=0.6459, include_large_r_jets=False, label="")

Reads the public ATLAS Open Data 13 TeV mini-ntuple format
(opendata.atlas.cern) -- object-level only, no tracks/vertices.

Requires: pip install "trace-hep[delphes]"
URL discovery requires: pip install "trace-hep[opendata]"
```

Loader reference (tracehep.io) 3/4

```
cms_opendata.load_event(path, index, tree_name="Events",
                        btag_cut=0.679, label="")
cms_opendata.load_events(path, indices, tree_name="Events",
                         btag_cut=0.679, label="")

Reads the public CMS Open Data reduced NanoAOD format
(opendata.cern.ch, AOD2NanoAODOutreachTool) -- 2012, 8 TeV, MC
and real collision data. path can be a local file OR an
https://opendata.cern.ch/... URL (streamed, not downloaded).

Requires: pip install "trace-hep[delphes]"
```

Loader reference (tracehep.io) 4/4

```
atlas_physlite.load_vertex_event(path, event_index,
         tree_name="CollectionTree", label="")
atlas_physlite.load_event_jets(path, event_index,
     tree_name="CollectionTree", jet_pt_min=20.0)
```

Reads ATLAS Open Data's 2024 research release (DAOD_PHYSLITE). Every vertex's real position/is_hs/is_pu is returned; track_indices is non-empty wherever PHYSLITE kept tracks -- nearly always the hard-scatter vertex, ~3.6% of pileup vertices (measured, not assumed). Jets carry no vertex association or truth. No per-track timing in this release.

Requires: pip install "trace-hep[delphes]"

Filter reference (tracehep.filters)

```
filter_event(event, *, jet_pt_min=None, jet_pt_max=None,
   jet_eta_min=None, jet_eta_max=None, track_pt_min=None,
 track_pt_max=None, track_eta_min=None, track_eta_max=None)

filter_vertex_event(vertex_event, *, track_pt_min=None,
 track_pt_max=None, track_eta_min=None, track_eta_max=None)

filter_jets(jets, *, pt_min=None, pt_max=None,
   eta_min=None, eta_max=None, btag_only=False)
filter_tracks(tracks, *, pt_min=None, pt_max=None,
              eta_min=None, eta_max=None)

All four return a filtered COPY -- the input is never modified.
```

Gallery reference (tracehep.gallery)

```
build_gallery(events, categories, plot_fn, output_path, *,
     title="TRACE Event Review", dpi=100, captions=None)

compare_pass_fail(results_a, results_b, *,
        name_a="algo1", name_b="algo2")
```

events/categories: {event_id: object}/{event_id: label} mappings.
plot_fn(event) must return a matplotlib or plotly figure.
dpi controls embedded-image size -- drop to 60-80 for hundreds-
thousands of events to keep the single HTML file responsive.

No new dependency -- pure stdlib (base64/html) plus matplotlib.

Viewer reference (tracehep.webapp)

```
trace-gui [--host 127.0.0.1] [--port 5057] [--no-browser]
python3 -m tracehep.webapp   (equivalent)

create_app() -- returns the Flask app undecorated, for embedding
or testing (e.g. app.test_client()) without starting a server.

Loader dropdown covers all six tracehep.io loaders, plus "Build
an event by hand" (no file -- type jets/leptons/photons/MET
directly). URL query params pre-fill the form and auto-render.

Requires: pip install "trace-hep[gui]" (adds Flask).
```

Troubleshooting

* ImportError: uproot required
  -> pip install "trace-hep[delphes]"

* ModuleNotFoundError: No module named tracehep (right after install)
  -> from TestPyPI you need BOTH --index-url and
  --extra-index-url (see Installation slides)

* KeyError from flat_ntuple loader
  -> the (run, event) pair is not present in that file

* A track's 3D origin looks wrong from a calo-timing ntuple
  -> that format has no per-track (x, y, z); origin defaults
  to the owning vertex's (x, y) plus the track's own z0

Versioning and updates

* Published package versions are immutable
-> updating requires bumping the version and re-uploading

* Typical update loop:
edit code in src/tracehep/
add/update a test
bump version in pyproject.toml and __init__.py
python3 -m build
twine upload --repository-url https://test.pypi.org/legacy/ dist/*

Get involved

Source: github.com/wasikatcern/TRACE-HEP
Install: pip install trace-hep (TestPyPI now; PyPI planned)
License: MIT
Tests: pytest -q (52 passing at v0.1.13)

Contact: wasikul.islam@cern.ch

Citing: accompanying paper citation to be added once posted.